DRIVE SYSTEM INSTRUCTIONS

GEK-65151A

for

DS 3 8 2 0 HTM

Directo-Matic/PLUS

HIGHWAY TEST MODULE USER MANUAL

General Electric Company
Drive Systems Department
Salem, Virginia U.S.A.

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

****
* *
* WARNING *
* *
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To avoid potential electrical shock and misoperation of this equipment, the HTM must be properly grounded through the three-prong power plug. Do not use two-prong "cheater plugs". Also ensure that the electrical outlet in use has been properly grounded.

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SECTION 1
GENERAL DESCRIPTION

1-1 SCOPE

This instruction manual describes the Highway Test Module (HTM) and covers its installation, operation, and maintenance.

1-2. PURPOSE AND USE

The HTM facilitates troubleshooting and maintenance of the Directo-Matic® Plus Control Signal Highway (CSH) (high-speed serial data link).

1-3. GENERAL DESCRIPTION (See Figure 1-1).

The HTM is a portable test set mounted in a hinged, aluminum case which opens to provide access to the front panel controls and indicators. The front panel is divided into three operating modes. There is also an array of common controls and indicators across the top of the panel. Test points are located below the I/O mode and slave mode section of the panel.

1-4. REFERENCE DATA

Size: length 21 inches (53.3 cm.)
width 17 inches (43.2 cm.)
depth 8 inches (20.3 cm.)

Weight: 33 pounds (15 kg.)

Operating Temperature Limits: 0°C to 50°C

Humidity: 5% to 95% relative humidity with no condensation.

Line Power: 95 VAC to 130 VAC (115 VAC NOMINAL)
1 Ampere load (nominal)
47 Hz to 440 Hz

Cables: power cable
transmit cable (RG179 coaxial)
receive cable (RG179 coaxial)
signal cable (48 conductor) to maintenance test connector.

Grounding: Unit must be grounded through power cord to operate.
Do not defeat ground plug.

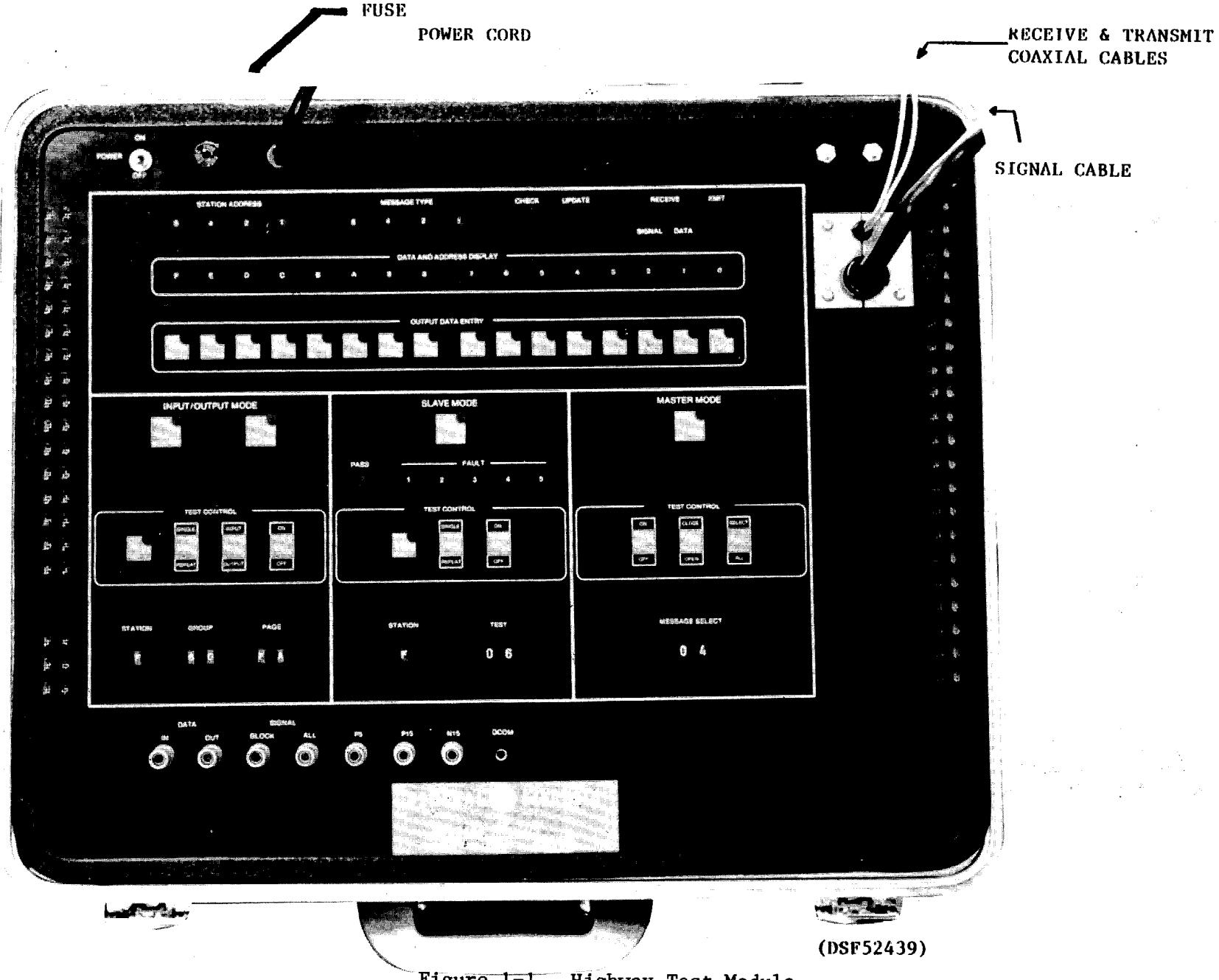


Figure 1-1. Highway Test Module

1-5. RELATED PUBLICATIONS

Refer to the instruction manual for the Installation and Maintenance Module (IMM). Familiarity with the IMM is helpful to the technician in conducting some of the tests described herein.

1-6. DESCRIPTION OF CONTROLS, INDICATORS, AND TEST POINTS

Table 1-1 describes all controls, indicators, and test points of the HTM. See Figure 1-1 for location of controls, indicators, and test points.

NOTE

Panel markings, control names and cable names actually appearing on the equipment are shown in upper-case letters throughout this publication.

Table 1-1. HTM Controls, Indicators, and Test Points

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
COMMON CONTROL AND INDICATORS	
POWER ON/OFF switch (toggle switch)	ON position applies 115 VAC to the HTM.
1.5 Ampere AC line fuse	Protects the HTM.
STATION ADDRESS indicators (four red light emitting diodes (LEDs))	Display the station address portion of the received CSH messages.
MESSAGE TYPE indicators (four red LEDs)	Display the four-bit message code of received CSH messages.
CHECK indicator (red LED)	Indicates that the received message had a valid Cyclic Redundancy Check (CRC) field.
UPDATE indicator (red LED)	Lights momentarily whenever new data is displayed.
RECEIVE SIGNAL indicator (red LED)	Indicates that the HTM senses modulation on its receive coaxial cable.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
COMMON CONTROL AND INDICATORS (Continued)	
RECEIVE DATA (red LED)	Indicates that the HTM senses messages on its receive coaxial cable which are other than Idle or Synchronizing messages.
XMIT indicator (red LED)	Lights momentarily whenever the HTM initiates transmission other than an Idle message.
DATA AND ADDRESS DISPLAY indicators (16 red LEDs)	Display the information field of received messages.
OUTPUT DATA ENTRY switches 0-F (16 momentary pushbuttons)	Control the data portion of HTM transmitted messages. Switches are alternate action and are back-lighted to indicate their current state (on = logic 1).
TEST POINTS	
DATA IN	Permits monitoring serial data stream received by the HTM.
DATA OUT	Permits monitoring serial data stream transmitted by the HTM.
SIGNAL BLOCK	Permits monitoring scope synchronizing signal which goes negative whenever the HTM indicates a transmission other than Idle.
SIGNAL ALL	Permits monitoring synchronizing signal which goes negative during the 3-bit sync field of all HTM messages.
P5	Permits monitoring HTM logic power (+5VDC).
P15	Permits monitoring HTM +15VDC supply voltage.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
TEST POINTS (Continued)	
N15	Permits monitoring HTM -15VDC supply voltage.
DCOM	Common connection for all test point measurements.
MASTER MODE	
MASTER MODE switch (momentary pushbutton with yellow LED)	Selects master station test mode. Associated yellow LED lights master mode when selected.
DEMAND ON/OFF switch (rocker switch)	ON position causes the HTM to transmit an interrupt to the master station.
LOOP OPEN/CLOSE switch (rocker switch)	OPEN position causes the HTM to simulate an open loop condition (broken coax or malfunctioning station).
DISPLAY SELECT/ALL (rocker switch)	SELECT position causes the display of only the message indicated by the thumbwheels. ALL position causes all received messages to be displayed.
MESSAGE SELECT switch (thumbwheel)	In normal operation selects the message to display out of a burst of messages. In self-test (ØØ), selects HTM switches to be tested.
SLAVE MODE	
SLAVE MODE switch (momentary pushbutton with yellow LED)	Selects slave test mode. Associated yellow LED lights when slave mode selected.
PASS LED (yellow LED)	Lights when test is successfully passed.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
SLAVE MODE (Continued)	
FAULT LEDs 1-5 (red LEDs)	One or more LEDs will normally light to indicate a malfunction. LED number helps fault isolation under some conditions.
TEST START switch (momentary pushbutton with red LED)	Initiate selected test. Associated red LED will light very briefly.
RUN SINGLE/REPEAT switch (rocker switch)	SINGLE position causes selected test to run once. REPEAT position causes selected test to run continuously.
FAULT STOP ON/OFF switch (rocker switch)	Aborts test upon occurrence of a failure when in ON position.
STATION switch (thumbwheel switch)	Selects desired address (1 to F) of slave station to be tested.
TEST switch (thumbwheel switch)	Selects desired slave station test.
I/O SIMULATION MODE	
SIMULATE switch (momentary pushbutton)	Selects I/O simulate mode of operation. Associated yellow LED lights when I/O simulation mode selected.
TEST switch (momentary pushbutton with red LED)	Initiates an input or output simulation.
MODE SINGLE/REPEAT switch (rocker switch)	Selects single or repetitive operation.
SELECT INPUT/OUTPUT switch (rocker switch)	Selects input or output operation.
PAGE ON/OFF switch (rocker switch)	Not used in I/O simulation mode.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
I/O SIMULATION MODE (Continued)	
STATION/GROUP/PAGE switches (thumbwheel switches)	Selects desired I/O address to be input (from) or output (to). If PAGE address is not used at the station, PAGE thumbwheel must be set at 00.
OUTPUT DATA ENTRY switches 0-F (16 momentary pushbuttons)	Enters data to be written to output.
I/O MONITOR MODE	
MONITOR switch (momentary pushbutton)	Selects monitor mode of operation. Associated yellow LED lights when monitor mode selected.
TEST switch (momentary pushbutton with red LED)	Enables single monitor when MONITOR SINGLE/REPEAT switch is in SINGLE position.
MODE SINGLE/REPEAT switch (rocker switch)	Selects either single or repeat (continuous) monitor mode operation. In REPEAT position, the selected I/O message is displayed each time it is detected. In SINGLE position, the I/O message is displayed once each time TEST switch is depressed.
SELECT INPUT/OUTPUT switch (rocker switch)	Selects input or output operation.
PAGE ON/OFF switch (rocker switch)	Disables PAGE thumbwheel switch. PAGE switch must be set to OFF position to monitor I/O which is not using the page function.
STATION/GROUP/PAGE switches (thumbwheel switches)	Selects desired I/O address (input or output) to be monitored. A page address can be monitored only if the PAGE switch is in the ON position.
OUTPUT DATA ENTRY switches 0-F (16 momentary pushbuttons)	Not used in I/O monitor mode.

1-7. MODES OF OPERATION

The HTM has three modes of operation as described in paragraphs 1-8 through 1-10. Only one mode can be active at a time; the other two modes are disabled by internal circuitry. Mode selection is accomplished by depressing then releasing the alternate action mode switch.

1-8. INPUT/OUTPUT MODE. The I/O mode provides for on-line monitoring of I/O data from an operating system. It also provides for off-line simulation of a Programmable Master Controller (PMC) read/write function (the HTM replaces the PMC for purposes of this test). These two functions are shown in Figure 1-2 and described in paragraphs 1-9 and 1-10.

1-9. Monitor Function. To monitor I/O data, the HTM RECEIVE coax is "T" connected to the CSH. The HTM searches the bit stream for input or output data (selected by the panel switches) and displays it on the STATION ADDRESS, MESSAGE TYPE and DATA AND ADDRESS DISPLAY indicators. The HTM must be connected downstream from the point where data enters the stream.

1-10. Simulate Function. The HTM may be used to simulate a master station. As such it can input or output data to a bypassed (isolated) slave station or to a system by replacing the PMC.

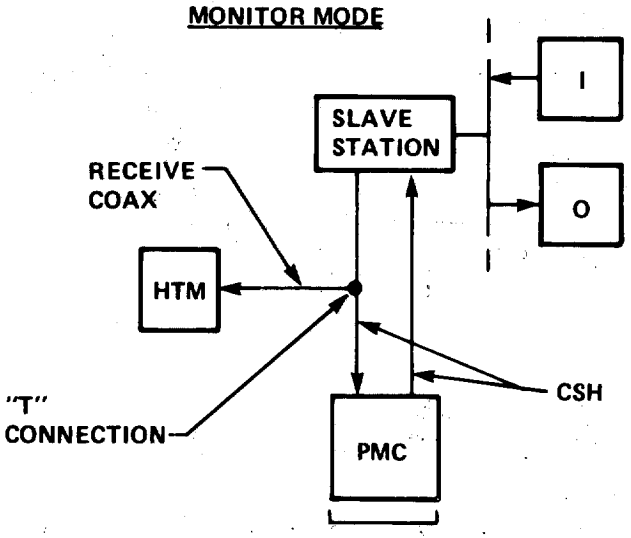


Figure 1-2. Input/Output Modes

1-11. SLAVE MODE. The slave mode provides for off-line test of equipment operating in the slave mode. This equipment can be either a:

- PMC operating in slave mode
- Slave station
- Group of slave stations in a system which may include a PMC operating in slave mode.

1-12. Slave Station Testing. In the slave mode, the HTM performs a series of tests on the slave station and provides a pass/fail indication of the results. In the event of a failure, the HTM provides identification of the most probable failed printed wiring board (PWB). These tests must be run in sequential order (by test number) for the test results to be valid. For proper operation of the HTM in slave station testing, and to prevent spurious outputs, the I/O terminal must be taken off line.

NOTE

Each PMC and slave station contain cards which provide communication with the CSH. These cards are referred to as the "CSH card complement."

1-13. System Testing. In the slave mode, the HTM replaces the PMC for system level installation and validation. Test software in the PMC computer supports checkout of the station CSH card complement. Successful completion of the hardware test (for a station on a loop) will validate approximately 60% of the station CSH PWB complement. None of the station's I/O capability is exercised. The master mode should also be used to thoroughly validate a PMC CSH card complement.

1-14. MASTER MODE. The master mode provides for:

- Off line test of a PMC operating in the master mode, and
- Identification of the most probable failed PWB.

PMC tests are controlled by test software resident in the PMC computer. These tests are comprised of a series of tests with go - no/go results reported by the computer.

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SECTION 2
OPERATING INSTRUCTIONS

2-1. PREPARATION FOR TEST

Prior to using the HTM, it should be tested using test 00 and test 01 instructions. These instructions appear in Section 4.

2-2 IDENTIFICATION OF PWB's. Throughout this publication, four-letter capitalized abbreviations such as HERA (the latter portion of GE catalog number DS3800HERA) are used to identify various types of PWB's.

2-3. HLXA PWB COAXIAL CONNECTORS. Figure 2-1 shows the location and name of each HLXA PWB coaxial connector. An HLXA PWB is located in each of the CSH card complements.

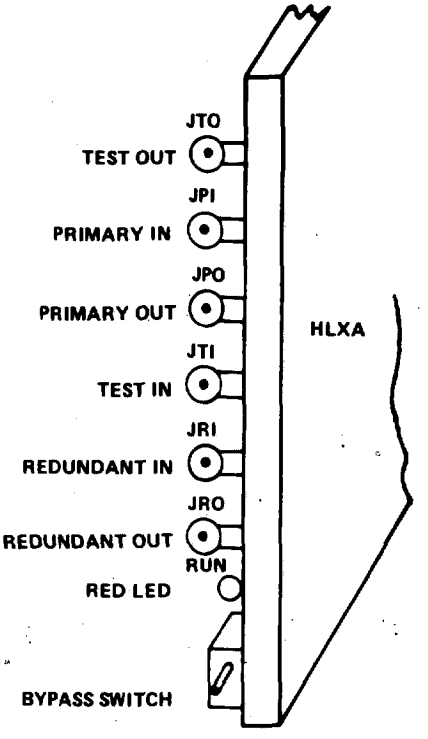


Figure 2-1. HLXA Coaxial Connectors

NOTE

To prevent inadvertent outputs to the CSH do not connect TX cable.

WARNING

THE HTM IS INTENDED FOR OFF LINE TESTING OF THE CSH, EXCEPT WHEN USING THE MONITOR MODE. WHEN TAKING A PMC, I/O TERMINAL OR SLAVE STATION OFF LINE, SAFEGUARDS SHOULD BE TAKEN TO PREVENT THE INADVERTENT STARTING OR STOPPING OF A PROCESS WHICH COULD RESULT IN INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT.

2-4. TEST INSTRUCTIONS

Perform test procedures as described in paragraphs 2-5 through 2-8.

2-5. I/O MONITOR

a. Connect the HTM as shown in Figure 2-2.

b. Momentarily depress the MONITOR switch. Observe that associated yellow LED lights.

c. Set SELECT INPUT/OUTPUT switch to INPUT or OUTPUT as desired.

d. Set STATION, GROUP and PAGE thumbwheels to the input or output address of interest. If page addressing is not used, set PAGE SELECT switch to OFF.

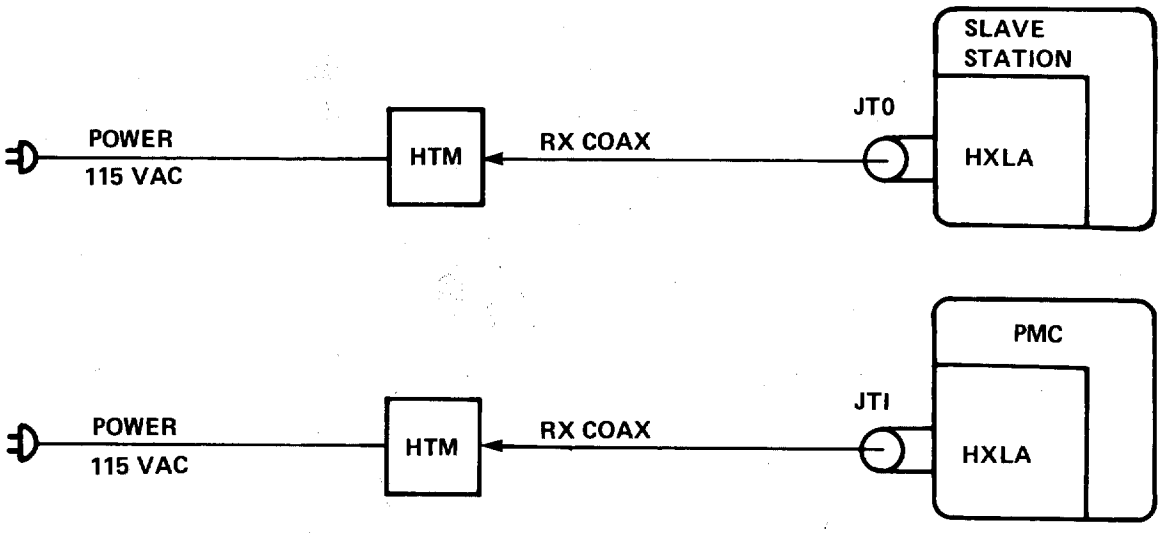


Figure 2-2. I/O Monitor Connections

e. Set MODE SINGLE/REPEAT switch to REPEAT if continuous monitoring is desired. If a single message is desired, set MODE switch SINGLE position, then depress TEST switch momentarily to obtain one message.

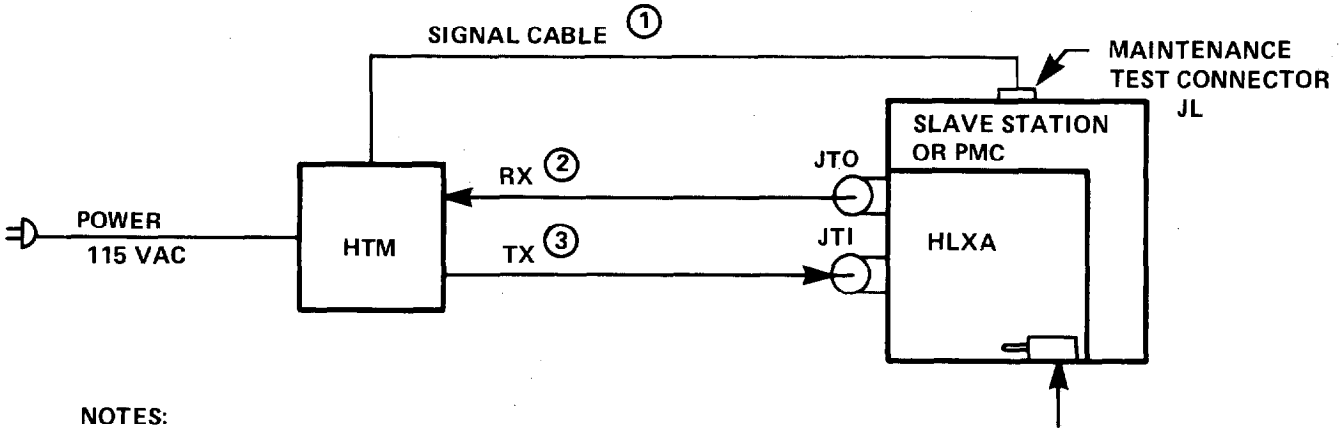
f. Observe that RECEIVE SIGNAL and RECEIVE DATA LED's light, indicating CSH activity.

g. Refer to system elementary diagrams for STATION, GROUP, and PAGE codes applicable to specific systems.

2-6. I/O SIMULATE

a. Set HLXA toggle switch to center position to bypass station.

b. Connect HTM as shown in Figure 2-3 to run station tests or as shown in Figure 2-4 to run system test.



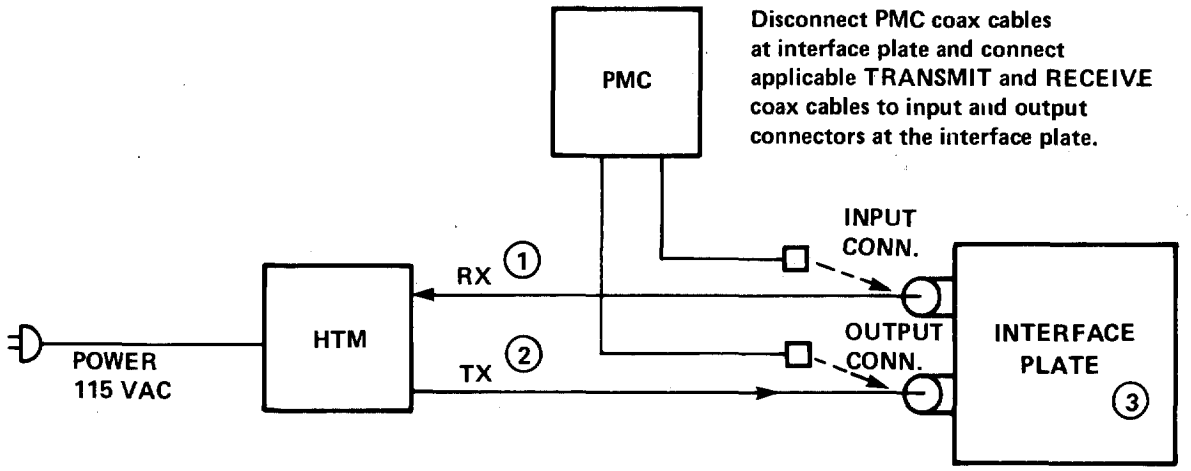
NOTES:

1 USED ONLY FOR SLAVE STATION TEST

2 RECEIVE COAXIAL CABLE

3 TRANSMIT COAXIAL CABLE

Figure 2-3. I/O Simulate or Slave Connections (Individual Station Tests - Only).



	PRIMARY CABLE		REDUNDANT CABLE	
LOOP	INPUT CONN	OUTPUT CONN	INPUT CONN	OUTPUT CONN
J	JAPI	JAPO	JARI	JARO
K	JBPI	JBPO	JBRI	JBRO
L	JCPI	JCPO	JCRI	JCRO

NOTES:

1. RECEIVE coaxial cable
2. TRANSMIT coaxial cable
3. The interface plate contains the coaxial connectors which interface customer cables with the PMC.

Figure 2-4. I/O Simulate or Slave Connections (Connections for Complete System Test-Only).

c. Set HTM switches as follows:

Switch	Position/Condition
POWER ON/OFF	ON
SIMULATE	Depress momentarily (associated yellow LED lights).
SELECT INPUT/OUTPUT	INPUT or OUTPUT as desired
STATION/GROUP/PAGE thumbwheels	Input or output address to be simulated. PAGE thumbwheel should be set to 00 on stations not using this function.

NOTE

The PAGE switch is not used for I/O simulation.

OUTPUT DATA ENTRY	Enter data to be written to the selected output address.
MODE SINGLE/REPEAT	REPEAT if a continuous simulation is desired.
	SINGLE if a single burst is desired. Then depress TEST switch momentarily to obtain a single burst of test data.

d. Return toggle switch on HLXA to normal setting.

2-7. SLAVE STATION TEST

a. Set toggle switch on HPBA and HLXA PWBs to center position (LEDs off) to put slave station off line.

b. Connect HTM as shown in Figure 2-3 to run station test and as shown in Figure 2-4 to run system test.

c. Set HTM switches as follows:

Switch	Position/Condition
POWER ON/OFF	ON
SLAVE MODE	Depress momentarily (associated yellow LED lights)

Switch	Position/Condition
STATION thumbwheel	Address of the station under test.
FAULT STOP ON/OFF	ON
RUN SINGLE/REPEAT	SINGLE
TEST	Desired test number (see Table 2-1) sequentially using only those checks applicable to the equipment to be tested (slave station, PMC, or slave stations in a system loop).
TEST START	Initiates test selected by TEST thumbwheel.

d. Observe slave mode front panel indicators:

(1) The yellow PASS LED lights to indicate successful completion of a test.

(2) The red FAULT LEDs 1-5 indicate test failure. See troubleshooting chart, Table 3-1, if FAULT LEDs light.

NOTE

Changing the TEST or STATION thumbwheels with the HTM running in the Repeat mode will cause faults to be displayed due to invalid switch positions.

2-8. MASTER STATION TEST

This procedure tests the master station in both the slave and master modes. The test uses the PMC software as well as the HTM. A basic knowledge of the IMM operation is therefore useful. The Appendix provides some operating information for the test program (CIAD), for those not already familiar with it. It is expected that this will normally be sufficient information to run these tests assuming that the software is loaded into the PMC and the IMM has been left in the normal operating mode.

a. Connect HTM as shown in Figure 2-5.

Table 2-1. Slave Mode Tests

TEST NUMBER	HARDWARE UNDER TEST: SLAVE STATION	HARDWARE UNDER TEST: PMC	HARDWARE UNDER TEST: SLAVE STATIONS IN SYSTEM	TEST DESCRIPTION	PROBABLE CAUSE OF FAULT
02	●	●		IDLE - CHECK RCV'D ADR/CODE & INFO = ZERO OR UNIQUE DEMAND	HLXA, HLXB
03	●	●	●	SYNCH - CHECK RCV'D ADR/CODE, INFO, AND CRC.	HLXA,HLXB IF FAULT #5 - ONLY, HLXB FAULTY
04 05 06 07	●	●	●	ACCEPT/TALK TEST DATA - CHECK RCV'D ADR/CODE, INFO. & CRC. TEST DATA { 04. '5555' 05. PANEL SWITCHES 06. 'AAAA' 07. '0000' TO 'FFFF'	SLAVE STATION - HLXB, HLSC IF FAULT #2 - ONLY, HLSC FAULTY MASTER STATION - HLXB, HLMB IF FAULT #2 - ONLY, HLMB FAULTY
08	●	●	●	TALK DATA - CHECK RCV'D ADR/CODE & CRC	HLXB
09	●	●	●	ACCEPT DATA - CHECK RCV'D ADR/CODE & CRC	
10 11 12	●	●	●	ACCEPT TALK & PAGE ADDRESS MESSAGES, CHECK RCV'D ADR/CODE, INFO, AND CRC	
13	●			ACC ADR PLUS 128 ACC DATA. CHECK GROUP ADR (INCM'T) ON DRIVE BUS.	HLSC/HLSD IF ALL TESTS IN THIS GROUP FAIL THE HLSC IS SUSPECT. IF A TEST FAILS AT SOME UNIQUE BIT PATTERN, THE HLSD IS SUSPECT
14	●			(ACC ADR/DATA) X128 - CHECK GROUP ADR (LOAD) ON DRIVE BUS.	
15	●			PAGE ADR X256, CHECK PAGE ADR ON DRIVE BUS.	
16	●			TALK ADR + 128 TALK DATA. CHECK GROUP ADR (INCM'T) ON DRIVE BUS	
17	●			TALK ADR X128, CHECK GROUP ADR (LOAD) ON DRIVE BUS.	
18	●			TALK DATA X 2^{15}. DATA FROM DRIVE BUS TO CSH.	HLSD/HLSC
19	●			ACC DATA X 2^{15}. DATA FROM CSH TO DRIVE BUS.	
20 21 22		●	PMC SLAVE	WRITE/READ CHECKS OF RAM ON HLMA, 20-INCREMENTAL DATA, 21 & 22 - CHECKERBOARD PATTERNS.	HLMB/HLMA FAILURES ON UNIQUE BIT PATTERNS INDICATE HLMA IS SUSPECT, OTHERWISE, THE HLMB.
23	●			CHECK STA. ERR. DETECTION, REPORT & CLEAR	HLSC/HLSD (SEE NOTE 5).
24	●			CHECK LINK ERR DETECTION, REPORT & CLEAR	SLAVE HLSA/HLSB
25	●			CHECK INTERRUPT REPORT AND ACKNOWLEDGE OF SAME	HLSD/HLSC
26	●			CHECK DRIVE BUS CONTROL LINES - HOLD/READ/WRITE	HLSD

NOTES

1. WHERE TWO PWB'S ARE LISTED, THE FIRST IS THE MOST PROBABLE CAUSE.
2. CABLE CONNECTED TO JG CONNECTOR ON SLAVE STATION (IF INSTALLED) SHOULD BE DISCONNECTED.
3. TEST NUMBERS ABOVE THOSE LISTED WILL ILLUMINATE ALL FIVE FAULT LIGHTS.
4. TESTS 00 & 01 ARE HTM SELF TESTS - SEE HTM MAINTENANCE INSTRUCTIONS.

5.THIS TEST WILL FAIL ON STATIONS IN WHICH PARITY IS INTENIONALLY DISABLED.

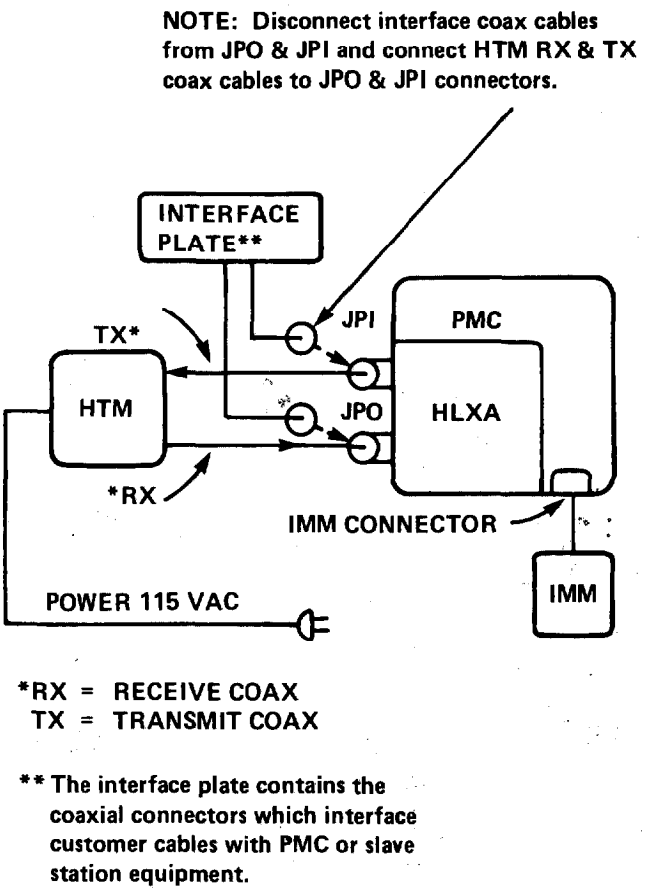


Figure 2-5. Master Mode Test Connections

b. Verify that when moving the MODE SELECTION switch on the PMC console to OFF LINE then back to STANDBY the indicator lights follow, or momentarily attempt to follow, to STANDBY indication before returning an OFF LINE indication. This demonstrates that the PMC hardware and software are operating normally.

c. Return the PMC console MODE SELECTION switch to OFF LINE.

d. Perform the following at the PMC.

(1) Turn off power supply PS5 located behind PMC console.

CAUTION

DO NOT TURN OFF POWER SWITCH ON PMC CONSOLE SINCE IT MAY REQUIRE SYSTEM TO BE RELOADED.

(2) Disconnect the JRI coaxial cable (if used), and disconnect the JPO and JPI coaxial cables from the HLXA. See Figure 2-6 for location of PWB's to ensure test is made on loop where problem is suspected.

(3) If HLXB address jumpers are set for "0", change to some other address per Figure 2-7. Record original address.

(4) Set HLXA toggle switch in bypass (center) position.

(5) Set HLMB jumper in "store" position.

e. Set HTM switches as follows:

Switch	Position/Condition
DEMAND ON/OFF	OFF
LOOP OPEN/CLOSE	CLOSE

f. Disable the stall timer on the PMC by jumpering TP1 to TP2 on HDIC PWB.

g. Turn PS5 POWER switch to ON.

h. Verify that top LED on HLMA is illuminated. This indicates that station is in master mode.

i. The IMM routine aborts any running program and turns on the CIAD program. The routine will be entered as indicated below, depending on the IMM status.

Keyboard	Display
RETURN key	IMM aborted
RETURN key	*
RUN, CIAD	* RUN, CIAD
RETURN key	#

j. The IMM test will put the master station in the slave mode. The top LED on the HLMA will light when the station is in the master mode.

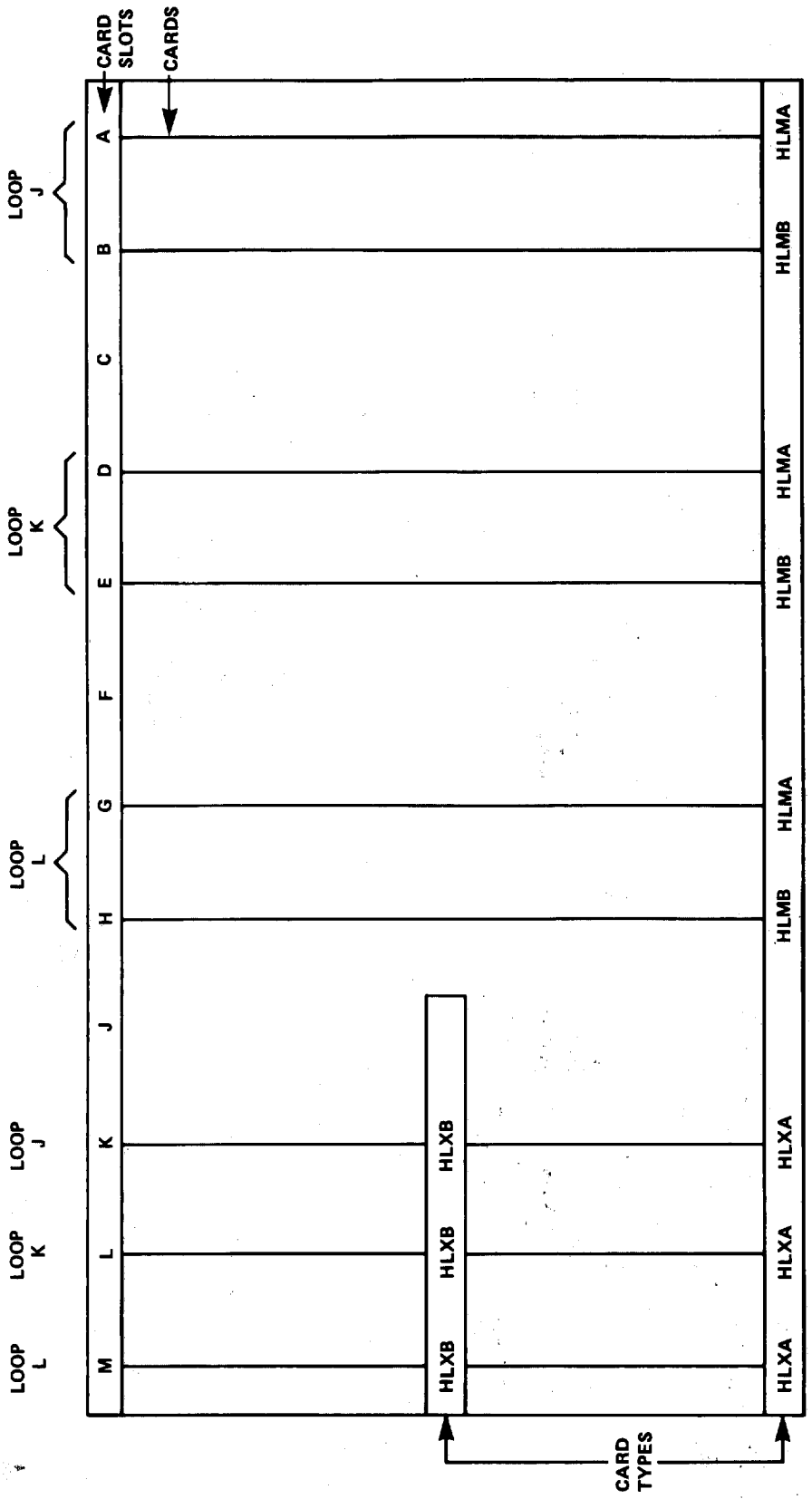


Figure 2-6. Master Station Layout

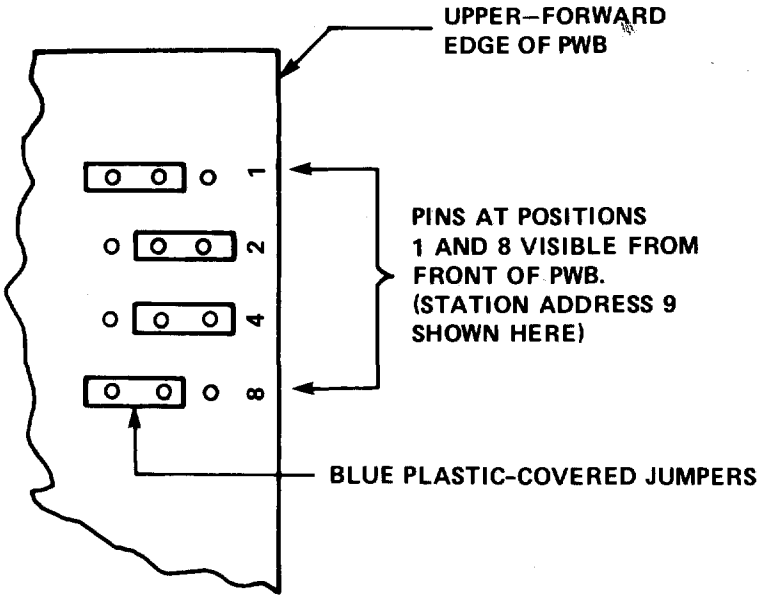


Figure 2-7. Jumpering for Station Address

Keyboard	Display	Failure
MTEST, X, 1	#MTEST, X, 1	
(X=J, K or L depending upon the loop under test).		
RETURN key	Status Error	HLMB
	Verify Error	HLMA
	#	Proceed to

k. If failure occurred during test, replace the suspected PWB, then turn PS5 off for 5 seconds and then back on. Verify that top HLMA LED is on and repeat test.

l. Verify that top HLMA LED is off at the PMC. If on, replace HLMB and repeat step j.

m. Select the slave mode by depressing the HTM SLAVE MODE switch.

n. Set the HTM station address thumbwheel to HLXB setting.

o. Run PMC tests per Table 2-1.

p. Select the master mode by depressing the HTM MASTER MODE switch.

q. Set HTM LOOP switch to CLOSE.

r. Set HTM DISPLAY switch to ALL.

s. Set HTM DEMAND switch to OFF.

t. Turn PMC PS5 off for 5 seconds, then back to on to return station to master mode.

u. Display IMM message as follows:

Keyboard	Display	Failure
MTEST, X, Z, C	# MTEST, X, Z, C	
RETURN key	Status Error	*HLMB/HLMA
	Interrupt Error	*HLMB/HLMA
	(Repeating Message)	Go to step r.

* Most probable failure.

r. Verify that the following IMM message is displayed repetitively - LINK CLOSED POLL DEMAND OFF.

w. Change the position of the HTM LOOP switch (OPEN/CLOSE) and verify that the IMM message corresponds. Leave switch in CLOSE position.

x. Change the position of the DEMAND switch (ON/OFF) and verify that the IMM message corresponds. Leave switch in OFF position.

y. If the IMM message fails to correspond with the switch positions in the previous two steps, the HLMB is most suspected of failure followed by the HLMA.

z. Set HTM LOOP switch to OPEN and verify that HTM indicators are as follows:

HTM Indicator	Condition
STATION ADDRESS	All ON
MESSAGE TYPE	All OFF
DATA AND ADDRESS DISPLAY	0-7 OFF 8-F ON

aa. If any indication in step z is incorrect, replace the HLMA.

ab. Set HTM LOOP switch to CLOSE.

ac. Display IMM message as follows:

Keyboard	Display	Failure
RETURN key	*	
BREAK, CIAD	Additional message followed by a #	
MTEST, X, Y	#MTEST, X, Y	

RETURN key #

X = loop number (J, K or L)

Y = 3 through 7 to run tests 3-7 sequentially. Do not enter each successive test unless a # is displayed with no error messages.

NOTE

If an error message occurs on any test, the probable failures are:

Status error - HLMB

Interrupt - HLMB

Verification - HLMA

ad. Depress ABORT key on IMM keyboard. Verify that display blanks.

ae. Reconnect HLMB and HLXB jumpers on the PMC to original positions.

af. Disconnect the HTM cables and reconnect the JRI, JPO, and JPI cables to the HLXA if disconnected in step 2.

af. Return the PMC MODE SELECTION switch to its original position. Return HLXA toggle switch to normal (UP) position.

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SECTION 3
THEORY OF OPERATION

3-1. PURPOSE OF HTM

The traffic controller with which the HTM is designed to be used, comprises a master station and up to 15 slave stations connected via a high-speed data link. A master station is the station at the head of each loop that formats and controls the traffic on the loop. The basic purposes of the HTM are described below:

a. The HTM determines whether a master station can be substituted for and checks if the data link works when the HTM is substituted for the master station. If a master station is suspected of being faulty, the HTM can be substituted for the master station and a check made to see if the data link works with the HTM.

b. The HTM can substitute for the entire data link, including the master station, and plugged into a slave station using the HTM in lieu of the entire data link. In other words, the slave station is taken out of the loop and connected to the HTM.

Refer to paragraph 1-7 for a description of HTM modes of operation.

3-2. TEST MESSAGE FORMAT

Two kinds of information are contained in the test message transmitted onto the data link by the HTM. This includes slave station addresses and data. Information received back from the slave stations includes 16 bits of data in the field of one message. The command field of the message contains information as to whom the message is destined for and the intent of the message. The HTM is capable of reconstructing all of the messages in the proper formats. There are approximately ten types of messages that involve transfer and status between master and slave.

3-3. DESCRIPTION

The HTM comprises eight printed circuit cards: HGPA, HGPB, HPIA, HHTA, HSQC, HLXA, HLXB, and HRRA. Cards HGPA and HSQC make up a test controller which generate all control signals for the HTM and establish format of test messages generated. Test, monitoring, and simulation functions performed by the HTM are selected by front panel switches. The commands selected are

interfaced to the A bus via the HPIA card. The A bus is in turn examined by card HGPA to determine what action to take. The messages generated by HGPA and HSQC are transmitted to the data link by transmitter card HHTA and transceiver cards HLXA/HLXB. In many modes of operation, the test message transmitted is also stored in a register file in the controller and a comparison is made of what was transmitted with what was received back on the data link. The register file is interfaced to bus A. The data coming back in is received by transceiver HLXA/HLXB. The received signal is clocked by receive PWB HRRA and placed on bus B. The A bus and the B bus are then compared in the controller to determine whether the test results are correct or incorrect. Usually a one for one comparison is made of the data going out with the data coming back in. However, in some cases, data received via the data link is different from that transmitted. In those cases, the information is not compared but displayed. Data to be displayed is interfaced to the appropriate front panel indicators via interface card HPIA.

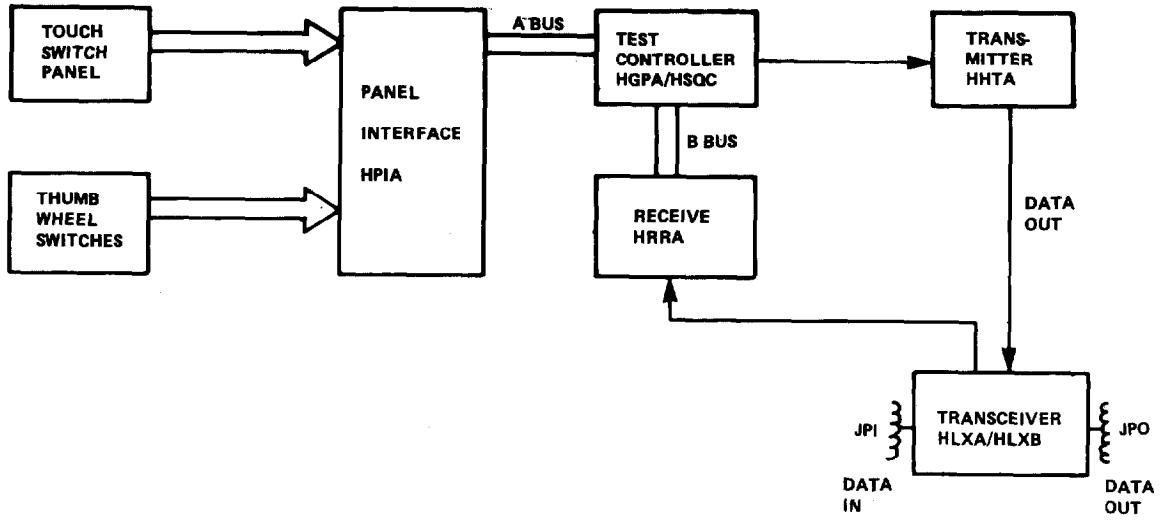


Figure 3-1. HTM Block Diagram

SECTION 4
HTM MAINTENANCE INSTRUCTIONS

4-1. PERIODIC MAINTENANCE

The HTM requires no periodic maintenance other than indicated in paragraphs 4-2 through 4-8.

4-2. CLEANING. Inspect the exterior of the HTM. Remove dust and other loose dirt with a clean soft cloth.

4-3. VISUAL INSPECTION. Inspect the HTM, prior to each operation, for obvious visual defects such as loose connectors, frayed cables, etc. When the equipment fails to operate properly, check switches and controls for correct setting.

4-4. POWER SUPPLY CHECKS. The DC power supply voltages should be checked periodically. To do so, turn on the POWER switch and check the power supply voltages at the HTM front panel test points. Voltages should be as follows:

Test Point	Required Voltage
P5	+4.75 to +5.25VDC
P15	+15 to +17.2VDC
N15	-15 to -17.2VDC

4-5. ADJUSTMENT OF +5VDC POWER SUPPLY P5. The +5VDC power supply, P5, requires periodic adjustment, if voltage measured at test point P5 is not within tolerance. This is the only adjustment on the HTM. P5 can be adjusted by opening the HTM and adjusting R40 on the power supply for +5V at test point P5.

4-6. SELF-TEST. The unit has a self-test (tests 00 and 01) which should be run prior to each use. Test 00 reads the switches and displays their value on the DATA AND ADDRESS DISPLAY for visual verification. Test 01 is a go/no-go check of transmit/receive capability. The self-test function verifies approximately 90% of the HTM circuitry. In the event that faults are detected during the self-test, fault isolation procedures are included as an extension to the 00 and 01 tests.

4-7. Test 00 - HTM Operator Device. Conduct test 00 as follows:

a. Momentarily depress the HTM SLAVE MODE switch (associated yellow LED lights).

b. Set RUN switch to REPEAT.

c. Set FAULT STOP switch to OFF.

d. Set TEST thumbwheel to 00.

e. Momentarily depress TEST START switch to initiate test. The PASS LED (yellow) and five FAULT LED's should illuminate (lamp test).

f. Refer to Table 5-1 and perform the following self test procedure:

Table 4-1. HTM Self-Test Switch/Display Matrix for Test 00

MESSAGE SELECT Thumbwheel Settings	HTM Front Panel Switches	HTM Displays Affected
00	OUTPUT DATA ENTRY Switches	DATA AND ADDRESS DISPLAY
01	Thumbwheel Switches PAGE GROUP	DATA AND ADDRESS DISPLAY
02	Slave Mode STATION Thumbwheel	STATION ADDRESS & DATA AND ADDRESS DISPLAY (1 of 16 addresses)
03 thru 99	MESSAGE SELECT Thumbwheel	STATION ADDRESS & MESSAGE TYPE

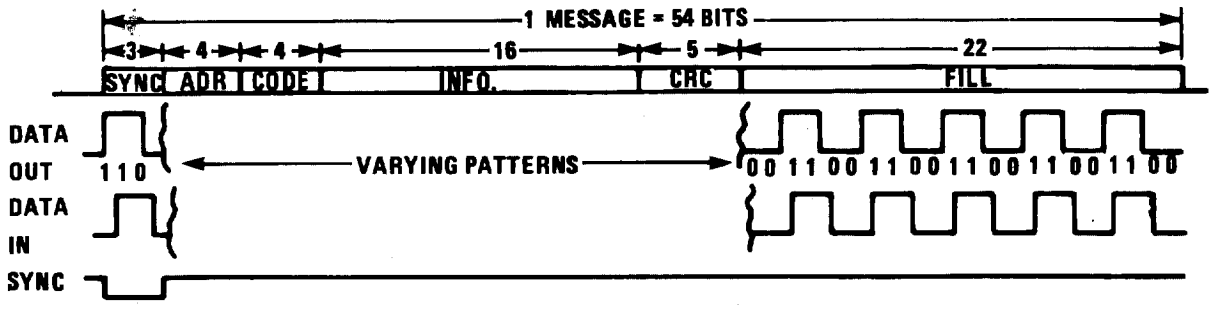


Figure 4-1. HTM Message Structure

4-9. TROUBLESHOOTING

Refer to Table 4-2 to troubleshoot the HTM. This table is based on the power supply checks of paragraph 4-4 and the self-test of paragraph 4-6.

Table 4-2. HTM Troubleshooting

SYMPTOM	PROBABLE CAUSE	REMEDY
All power supply voltages are OV	Defective fuse or power supply.	Replace the 1.5 ampere fuse on the front panel. If the fuse blows on replacement, replace the power supply.
While performing test 00, an operating mode cannot be selected on a test run.	Faulty HSQC or HGPA card.	Replace defective card.
While performing test 00, a single switch does not function.	Faulty switch or HPIA card.	Replace faulty switch or card.
While performing test 00, one display bit bad (for all switches).	Faulty HPIA card.	Replace card.
While performing test 00, alternate action or back lighting of OUTPUT DATA ENTRY switches bad.	Faulty HPIA card.	Replace HPIA card.

Table 4-2. HTM Troubleshooting Chart (Continued)

SYMPTOM	PROBABLE	REMEDY
While performing test 00, STATION thumbwheel is ok but 1 of 16 decoded outputs of DATA AND ADDRESS DISPLAY bad.	Faulty HPIA card.	Replace HPIA card.
While performing test 01, oscilloscope cannot be synchronized.	Faulty HHTA card.	Replace HHTA card.
While performing test 01, no data out (sweep only), or sync fill pattern not as shown in patterns in Figure 4-1.	Faulty HHTA card.	Replace HHTA card.
While performing test 01, data in does not follow data out as shown in Figure 4-1.	Faulty HLXA or HLXB card.	Replace HLXA and HLXB cards.
While performing test 01, data in/out patterns do not appear as shown in Figure 4-1.	Faulty HGPB or HHRA card.	Replace HGPB and HHRA cards.

4-10. DISASSEMBLY INSTRUCTIONS

WARNING

ONLY THOSE PERSONNEL QUALIFIED TO SERVICE ELECTRONIC EQUIPMENT SHOULD ATTEMPT TO DISASSEMBLE THE HTM. THE APPLICATION OF POWER TO THE HTM, WHILE DISASSEMBLED, PLACES LINE POWER ON EXPOSED TERMINALS. THEREFORE, PRECAUTION SHOULD BE TAKEN TO AVOID A SHOCK HAZARD. PRIOR TO REMOVING THE HTM FRONT PANEL, DISCONNECT POWER PLUG TO AVOID POSSIBILITY OF SHOCK HAZARD ON TERMINAL WHICH WILL BE EXPOSED.

4-11. REMOVAL OF HTM FRONT PANEL. Remove the HTM front panel as follows:

a. Remove top cover of HTM case by disengaging hinge.

b. Carefully lay HTM on flat working surface with HTM front panel down.

c. Remove 6 screws in bottom of aluminum case; lift case upward freeing internal assemblies attached to front panel.

4-12. REMOVAL OF POWER SUPPLY. Remove the power supply as follows:

a. With carrying case lying flat (power supply at bottom) disconnect connector at end of power supply.

b. Remove two screws adjacent to carrying handle; remove power supply.

4-13. REMOVAL OF THUMBWHEELS. Remove the thumbwheels as follows:

a. Remove connector by carefully pulling it away from back side of thumbwheel switch.

b. Push thumbwheel straight through and free of the HTM front panel.

4-14. REMOVAL OF FAN. Remove the fan as follows:

a. Remove two quick disconnect wire connectors.

b. While supporting fan motor, remove four screws and lay fan aside.

4-15. REMOVAL OF COAXIAL CABLES AND SIGNAL CABLE. Remove the coaxial cables and signal cable as follows:

a. Disconnect cable connectors from board edge.

b. Remove four screws from metal plate on HTM front panel through which the cables are routed, remove cables.

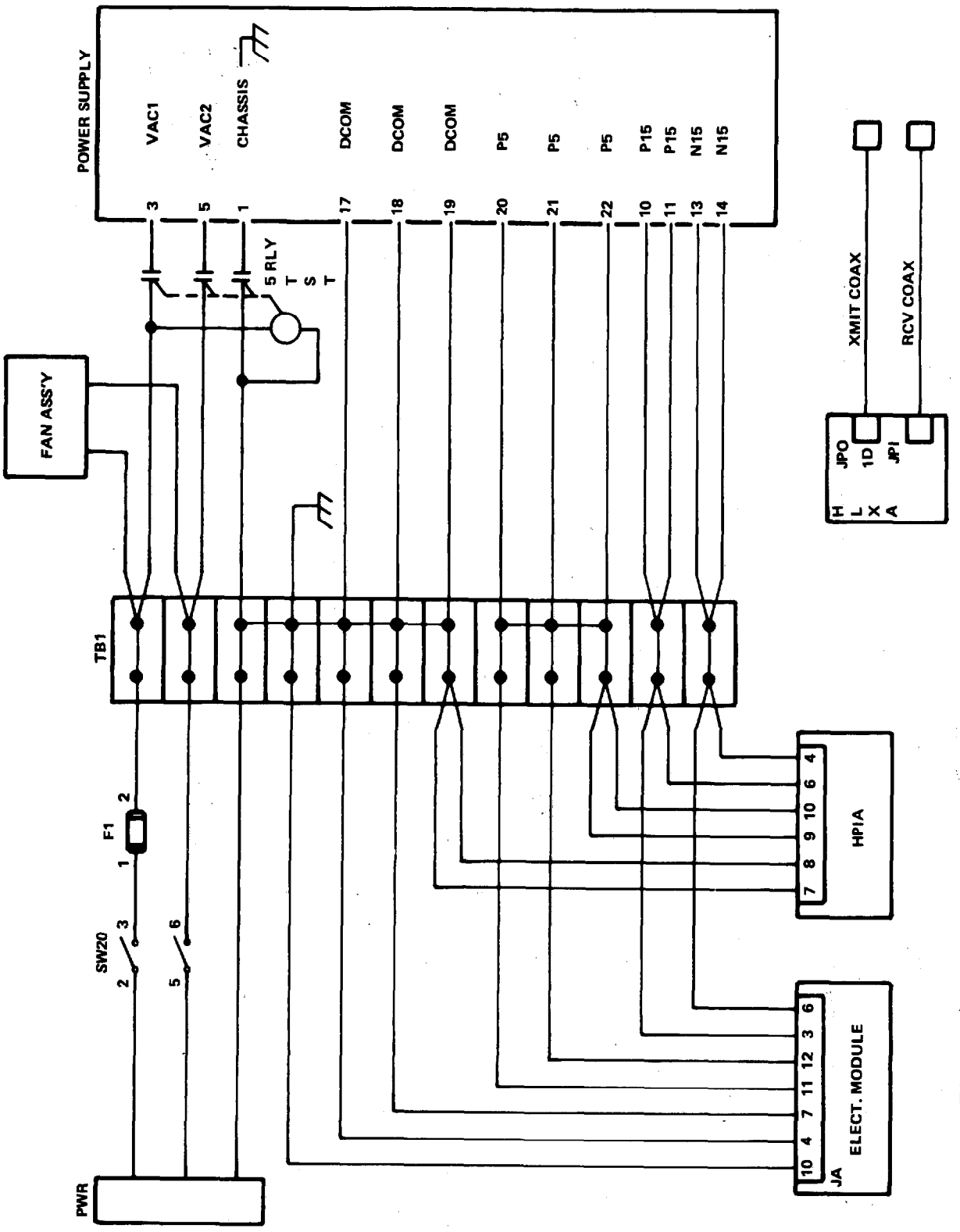


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 1 of 4)

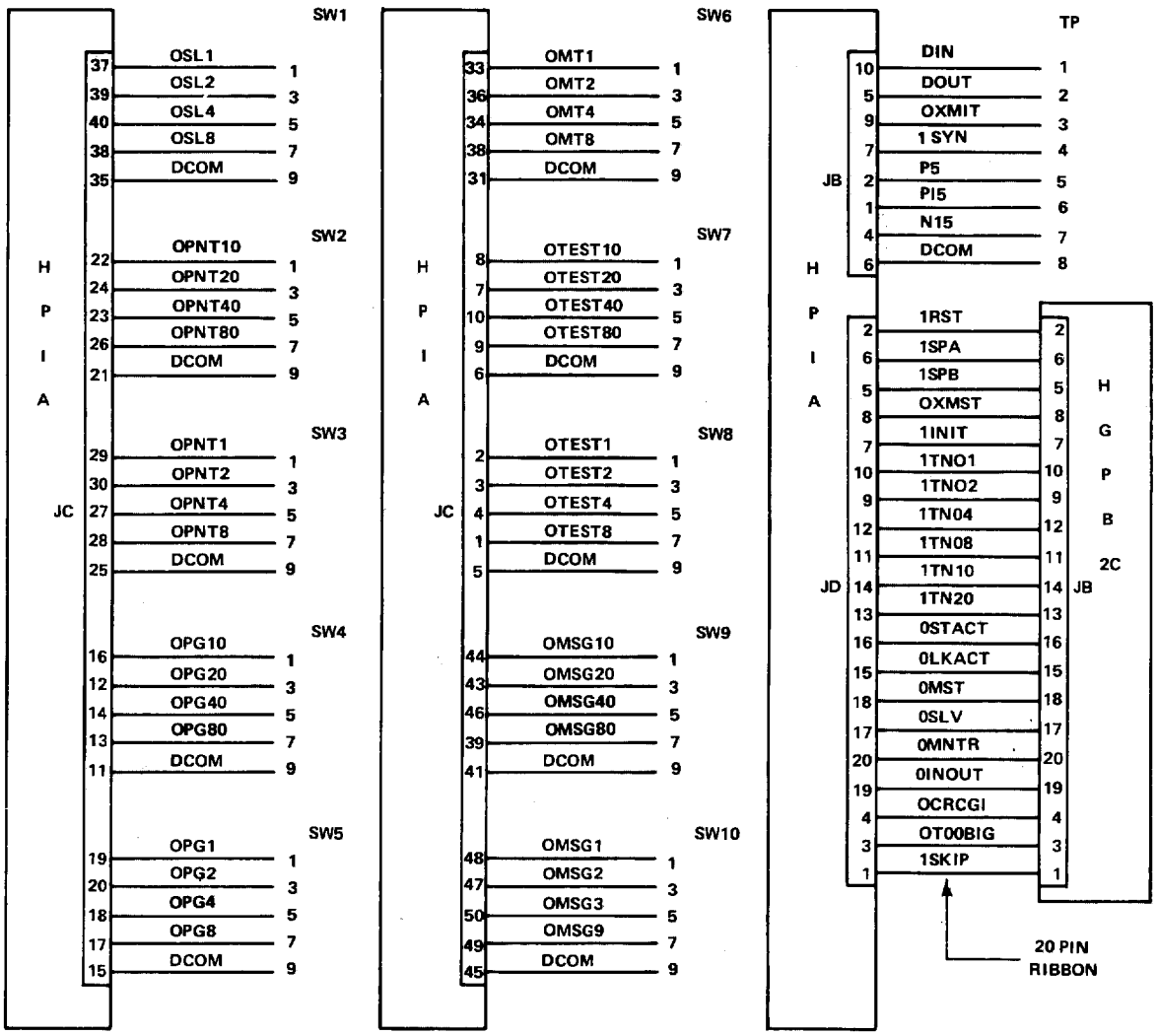


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 2 of 4)

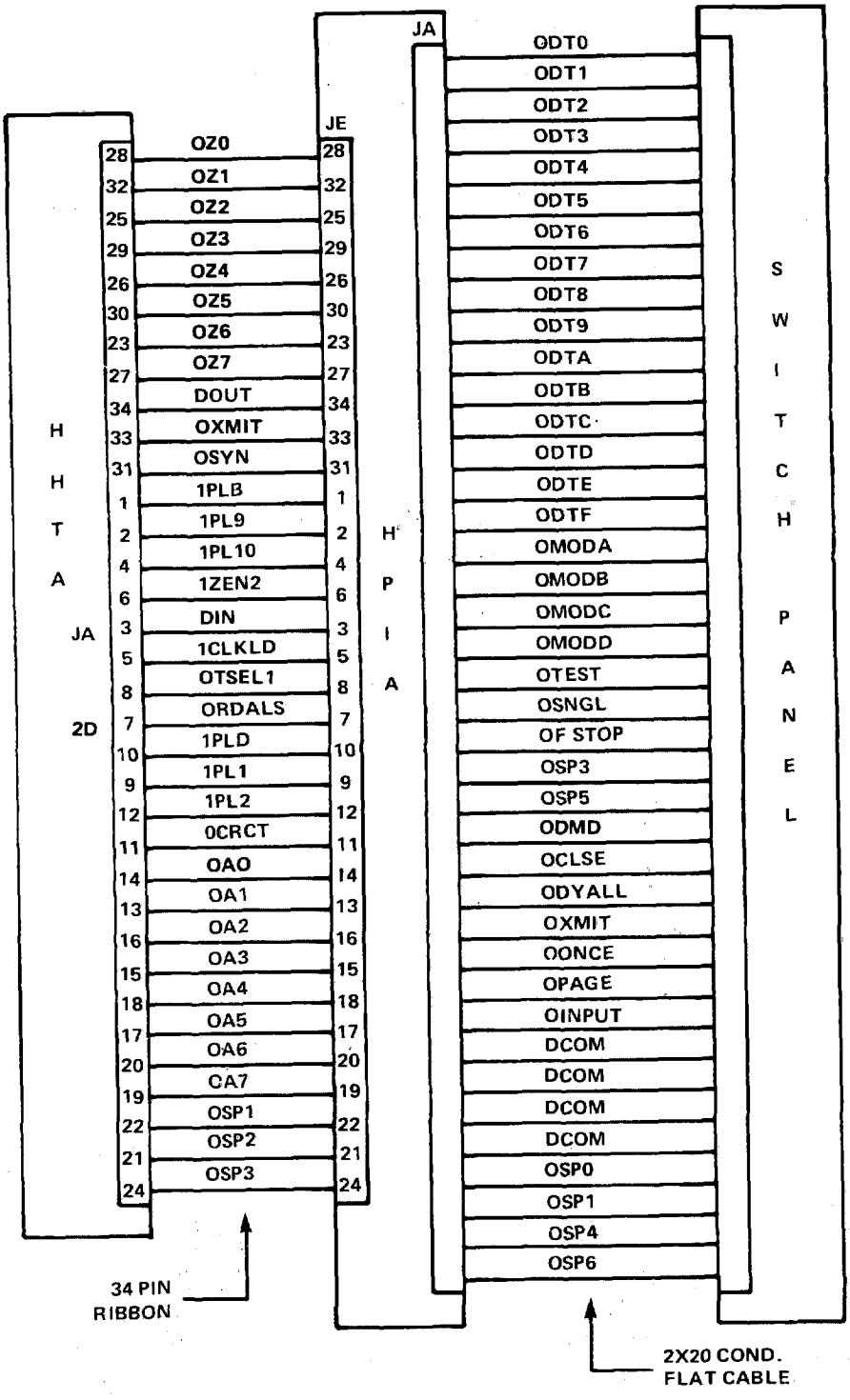


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 3 of 4)

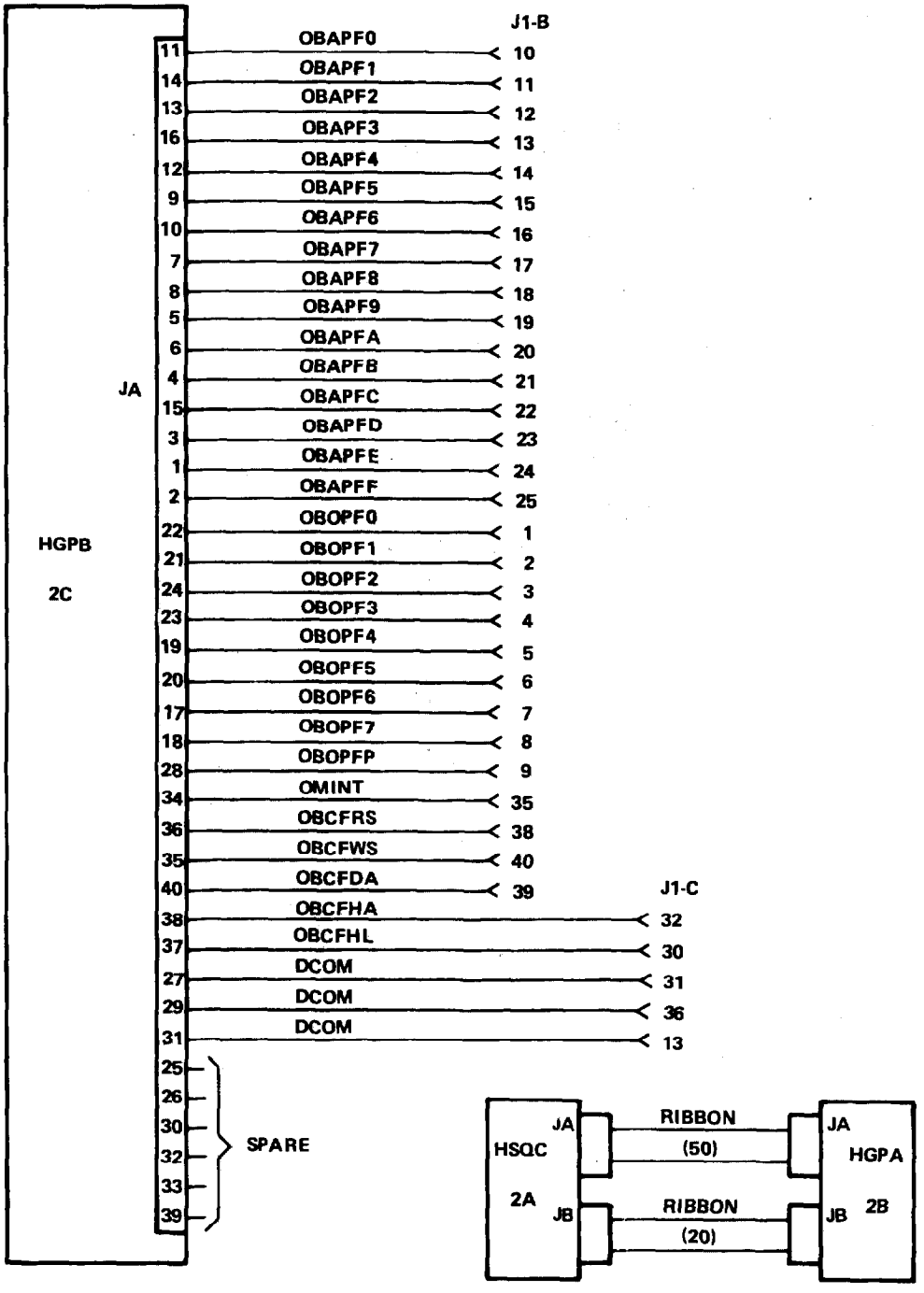


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 4 of 4)

SECTION 5
REPLACEMENT PARTS

Table 5-1 is a list of recommended spare parts for the HTM.

Table 5-1. HTM Spare Parts

NOMENCLATURE	CATALOG NUMBER
CARD	DS3800HHRA
CARD	DS3800HLXB
CARD	DS3800HLXA
CARD	DS3800HSQC
CARD	DS3800HGPA
CARD	DS3800HGPB
CARD	DS3800HHTA
FAN	218A4491P5
THUMBWHEEL SWITCH	218A4902P1
THUMBWHEEL SWITCH	218A4902P2
PANEL INTERFACE PWB	DS3800HPIA
POWER SUPPLY	218A4924P1
RIBBON CABLE	DS3815RCC1BL014
RIBBON CABLE	DS3815RCG1BL012
RIBBON CABLE	DS3815RCL1BL003
RIBBON CABLE	DS3815RCC1BL003
SIGNAL CABLE ASM	277A6574G1
COAXIAL CABLE ASM	277A6575G1
COAXIAL CABLE ASM	277A6575G2
FUSE 1.5 AMP	573A392P12
POWER SWITCH	68A7336P11
POWER INTERRUPT RELAY	218A4991P1

APPENDIX

TEST PROGRAM GENERAL INFORMATION

There are two programs which can be run on the IMM; the IMM program and the CIAD program for special tests

IMM PROGRAM

When the IMM program is running, ENTER COMMAND will appear as a prompt on the CRT. This indicates the program is ready for the next input.

CIAD PROGRAM

When the CIAD program is running, the prompt character is a #. The CIAD program is called by entering RUN, CIAD. The CIAD test is selected by entering MTEST, X, Y, C, F, where:

X = loop under test (J, K or L).

Y = test number.

C = continuous and is optional. This causes the test to run repetitively.

F = run continuous without stopping for errors and is optional. This is used for more sophisticated methods using an oscilloscope or logic analyzer.

TERMINATING A TEST

To terminate a test running in continuous mode, depress RETURN key and then enter BREAK, CIAD followed by a RETURN. The IMM will display remaining messages which have been stacked up followed by a # prompt. To terminate the CIAD program, enter ABORT following a # prompt character followed by a RETURN.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

DRIVE SYSTEM INSTRUCTIONS

GEK-65151A

for

DS 3 8 2 0 HTM

Directo-Matic/PLUS

HIGHWAY TEST MODULE USER MANUAL

General Electric Company
Drive Systems Department
Salem, Virginia U.S.A.

GENERAL ELECTRIC

GEK-65151A

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

****
* *
* WARNING *
* *
****

To avoid potential electrical shock and misoperation of this equipment, the HTM must be properly grounded through the three-prong power plug. Do not use two-prong "cheater plugs". Also ensure that the electrical outlet in use has been properly grounded.

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SECTION 1
GENERAL DESCRIPTION

1-1 SCOPE

This instruction manual describes the Highway Test Module (HTM) and covers its installation, operation, and maintenance.

1-2. PURPOSE AND USE

The HTM facilitates troubleshooting and maintenance of the Directo-Matic® Plus Control Signal Highway (CSH) (high-speed serial data link).

1-3. GENERAL DESCRIPTION (See Figure 1-1).

The HTM is a portable test set mounted in a hinged, aluminum case which opens to provide access to the front panel controls and indicators. The front panel is divided into three operating modes. There is also an array of common controls and indicators across the top of the panel. Test points are located below the I/O mode and slave mode section of the panel.

1-4. REFERENCE DATA

Size: length 21 inches (53.3 cm.)
width 17 inches (43.2 cm.)
depth 8 inches (20.3 cm.)

Weight: 33 pounds (15 kg.)

Operating Temperature Limits: 0°C to 50°C

Humidity: 5% to 95% relative humidity with no condensation.

Line Power: 95 VAC to 130 VAC (115 VAC NOMINAL)
1 Ampere load (nominal)
47 Hz to 440 Hz

Cables: power cable
transmit cable (RG179 coaxial)
receive cable (RG179 coaxial)
signal cable (48 conductor) to maintenance test connector.

Grounding: Unit must be grounded through power cord to operate.
Do not defeat ground plug.

1-5. RELATED PUBLICATIONS

Refer to the instruction manual for the Installation and Maintenance Module (IMM). Familiarity with the IMM is helpful to the technician in conducting some of the tests described herein.

1-6. DESCRIPTION OF CONTROLS, INDICATORS, AND TEST POINTS

Table 1-1 describes all controls, indicators, and test points of the HTM. See Figure 1-1 for location of controls, indicators, and test points.

NOTE

Panel markings, control names and cable names actually appearing on the equipment are shown in upper-case letters throughout this publication.

Table 1-1. HTM Controls, Indicators, and Test Points

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
COMMON CONTROL AND INDICATORS	
POWER ON/OFF switch (toggle switch)	ON position applies 115 VAC to the HTM.
1.5 Ampere AC line fuse	Protects the HTM.
STATION ADDRESS indicators (four red light emitting diodes (LEDs))	Display the station address portion of the received CSH messages.
MESSAGE TYPE indicators (four red LEDs)	Display the four-bit message code of received CSH messages.
CHECK indicator (red LED)	Indicates that the received message had a valid Cyclic Redundancy Check (CRC) field.
UPDATE indicator (red LED)	Lights momentarily whenever new data is displayed.
RECEIVE SIGNAL indicator (red LED)	Indicates that the HTM senses modulation on its receive coaxial cable.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
TEST POINTS (Continued)	
N15	Permits monitoring HTM -15VDC supply voltage.
DCOM	Common connection for all test point measurements.
MASTER MODE	
MASTER MODE switch (momentary pushbutton with yellow LED)	Selects master station test mode. Associated yellow LED lights master mode when selected.
DEMAND ON/OFF switch (rocker switch)	ON position causes the HTM to transmit an interrupt to the master station.
LOOP OPEN/CLOSE switch (rocker switch)	OPEN position causes the HTM to simulate an open loop condition (broken coax or malfunctioning station).
DISPLAY SELECT/ALL (rocker switch)	SELECT position causes the display of only the message indicated by the thumbwheels. ALL position causes all received messages to be displayed.
MESSAGE SELECT switch (thumbwheel)	In normal operation selects the message to display out of a burst of messages. In self-test (ØØ), selects HTM switches to be tested.
SLAVE MODE	
SLAVE MODE switch (momentary pushbutton with yellow LED)	Selects slave test mode. Associated yellow LED lights when slave mode selected.
PASS LED (yellow LED)	Lights when test is successfully passed.

Table 1-1. HTM Controls, Indicators, and Test Points (Continued)

CONTROL, INDICATOR, OR TEST POINT	FUNCTION
I/O SIMULATION MODE (Continued)	
STATION/GROUP/PAGE switches (thumbwheel switches)	Selects desired I/O address to be input (from) or output (to). If PAGE address is not used at the station, PAGE thumbwheel must be set at 00.
OUTPUT DATA ENTRY switches 0-F (16 momentary pushbuttons)	Enters data to be written to output.
I/O MONITOR MODE	
MONITOR switch (momentary pushbutton)	Selects monitor mode of operation. Associated yellow LED lights when monitor mode selected.
TEST switch (momentary pushbutton with red LED)	Enables single monitor when MONITOR SINGLE/REPEAT switch is in SINGLE position.
MODE SINGLE/REPEAT switch (rocker switch)	Selects either single or repeat (continuous) monitor mode operation. In REPEAT position, the selected I/O message is displayed each time it is detected. In SINGLE position, the I/O message is displayed once each time TEST switch is depressed.
SELECT INPUT/OUTPUT switch (rocker switch)	Selects input or output operation.
PAGE ON/OFF switch (rocker switch)	Disables PAGE thumbwheel switch. PAGE switch must be set to OFF position to monitor I/O which is not using the page function.
STATION/GROUP/PAGE switches (thumbwheel switches)	Selects desired I/O address (input or output) to be monitored. A page address can be monitored only if the PAGE switch is in the ON position.
OUTPUT DATA ENTRY switches 0-F (16 momentary pushbuttons)	Not used in I/O monitor mode.

1-11. SLAVE MODE. The slave mode provides for off-line test of equipment operating in the slave mode. This equipment can be either a:

- PMC operating in slave mode
- Slave station
- Group of slave stations in a system which may include a PMC operating in slave mode.

1-12. Slave Station Testing. In the slave mode, the HTM performs a series of tests on the slave station and provides a pass/fail indication of the results. In the event of a failure, the HTM provides identification of the most probable failed printed wiring board (PWB). These tests must be run in sequential order (by test number) for the test results to be valid. For proper operation of the HTM in slave station testing, and to prevent spurious outputs, the I/O terminal must be taken off line.

NOTE

Each PMC and slave station contain cards which provide communication with the CSH. These cards are referred to as the "CSH card complement."

1-13. System Testing. In the slave mode, the HTM replaces the PMC for system level installation and validation. Test software in the PMC computer supports checkout of the station CSH card complement. Successful completion of the hardware test (for a station on a loop) will validate approximately 60% of the station CSH PWB complement. None of the station's I/O capability is exercised. The master mode should also be used to thoroughly validate a PMC CSH card complement.

1-14. MASTER MODE. The master mode provides for:

- Off line test of a PMC operating in the master mode, and
- Identification of the most probable failed PWB.

PMC tests are controlled by test software resident in the PMC computer. These tests are comprised of a series of tests with go - no/go results reported by the computer.

SECTION 2
OPERATING INSTRUCTIONS

2-1. PREPARATION FOR TEST

Prior to using the HTM, it should be tested using test 00 and test 01 instructions. These instructions appear in Section 4.

2-2 IDENTIFICATION OF PWB's. Throughout this publication, four-letter capitalized abbreviations such as HERA (the latter portion of GE catalog number DS3800HERA) are used to identify various types of PWB's.

2-3. HLXA PWB COAXIAL CONNECTORS. Figure 2-1 shows the location and name of each HLXA PWB coaxial connector. An HLXA PWB is located in each of the CSH card complements.

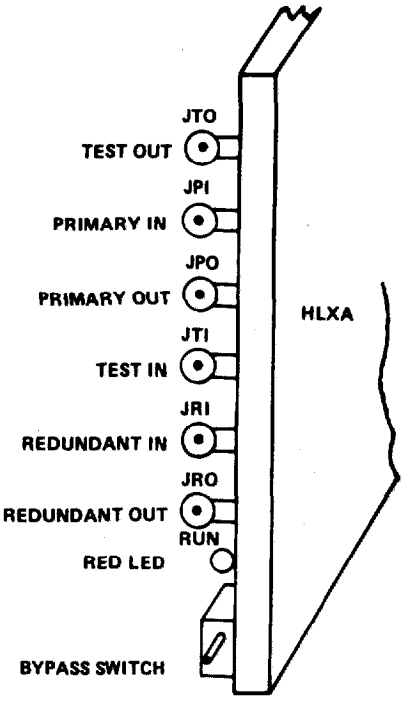


Figure 2-1. HLXA Coaxial Connectors

NOTE

To prevent inadvertent outputs to the CSH do not connect TX cable.

e. Set MODE SINGLE/REPEAT switch to REPEAT if continuous monitoring is desired. If a single message is desired, set MODE switch SINGLE position, then depress TEST switch momentarily to obtain one message.

f. Observe that RECEIVE SIGNAL and RECEIVE DATA LED's light, indicating CSH activity.

g. Refer to system elementary diagrams for STATION, GROUP, and PAGE codes applicable to specific systems.

2-6. I/O SIMULATE

a. Set HLXA toggle switch to center position to bypass station.

b. Connect HTM as shown in Figure 2-3 to run station tests or as shown in Figure 2-4 to run system test.

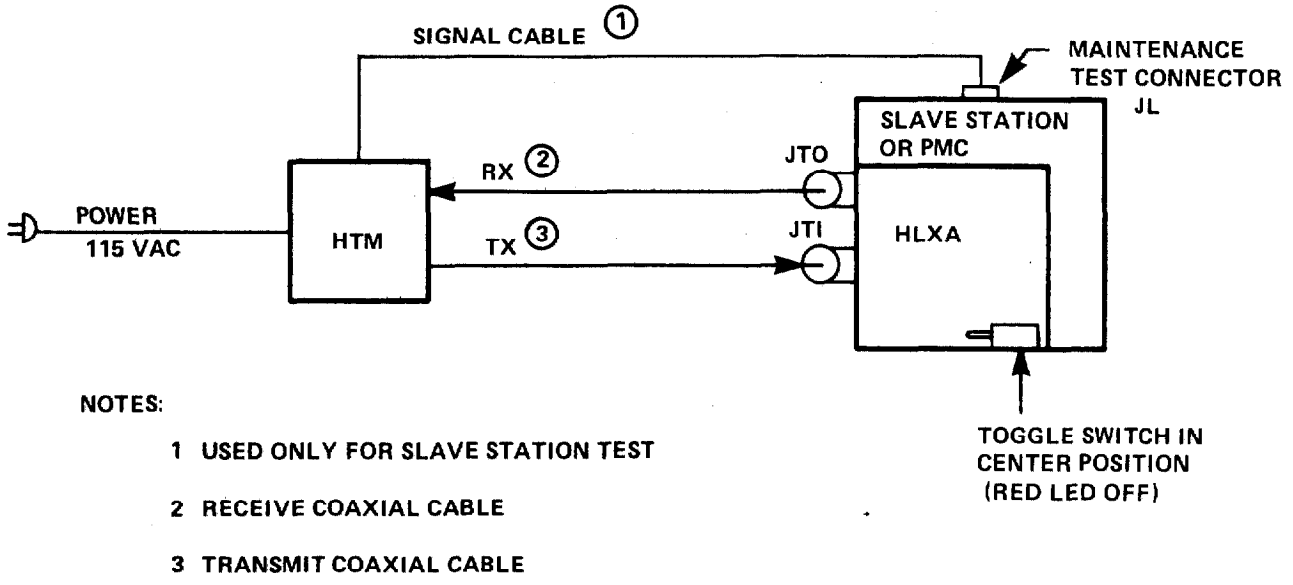


Figure 2-3. I/O Simulate or Slave Connections (Individual Station Tests - Only).

c. Set HTM switches as follows:

Switch	Position/Condition
POWER ON/OFF	ON
SIMULATE	Depress momentarily (associated yellow LED lights).
SELECT INPUT/OUTPUT	INPUT or OUTPUT as desired
STATION/GROUP/PAGE thumbwheels	Input or output address to be simulated. PAGE thumbwheel should be set to 00 on stations not using this function.

NOTE

The PAGE switch is not used for I/O simulation.

OUTPUT DATA ENTRY	Enter data to be written to the selected output address.
MODE SINGLE/REPEAT	REPEAT if a continuous simulation is desired. SINGLE if a single burst is desired. Then depress TEST switch momentarily to obtain a single burst of test data.

d. Return toggle switch on HLXA to normal setting.

2-7. SLAVE STATION TEST

a. Set toggle switch on HPBA and HLXA PWBs to center position (LEDs off) to put slave station off line.

b. Connect HTM as shown in Figure 2-3 to run station test and as shown in Figure 2-4 to run system test.

c. Set HTM switches as follows:

Switch	Position/Condition
POWER ON/OFF	ON
SLAVE MODE	Depress momentarily (associated yellow LED lights)

Table 2-1. Slave Mode Tests

TEST NUMBER	HARDWARE UNDER TEST: SLAVE STATION	HARDWARE UNDER TEST: PMC	HARDWARE UNDER TEST: SLAVE STATIONS IN SYSTEM	TEST DESCRIPTION	PROBABLE CAUSE OF FAULT
02	●	●		IDLE - CHECK RCV'D ADR/CODE & INFO = ZERO OR UNIQUE DEMAND	HLXA, HLXB
03	●	●	●	SYNCH - CHECK RCV'D ADR/CODE, INFO, AND CRC.	HLXA,HLXB IF FAULT #5 - ONLY, HLXB FAULTY
04 05 06 07	●	●	●	ACCEPT/TALK TEST DATA - CHECK RCV'D ADR/CODE, INFO. & CRC. TEST DATA { 04. '5555'; 05. PANEL SWITCHES; 06. 'AAAA'; 07. '0000' TO 'FFFF'	SLAVE STATION - HLXB, HLSC IF FAULT #2 - ONLY, HLSC FAULTY MASTER STATION - HLXB, HLMB IF FAULT #2 - ONLY, HLMB FAULTY
08	●	●	●	TALK DATA - CHECK RCV'D ADR/CODE & CRC	HLXB
09	●	●	●	ACCEPT DATA - CHECK RCV'D ADR/CODE & CRC	
10 11 12	●	●	●	ACCEPT TALK & PAGE ADDRESS MESSAGES, CHECK RCV'D ADR/CODE, INFO, AND CRC	
13	●			ACC ADR PLUS 128 ACC DATA. CHECK GROUP ADR (INCM'T) ON DRIVE BUS.	HLSC/HLSD IF ALL TESTS IN THIS GROUP FAIL THE HLSC IS SUSPECT. IF A TEST FAILS AT SOME UNIQUE BIT PATTERN, THE HLSD IS SUSPECT
14	●			(ACC ADR/DATA) X128 - CHECK GROUP ADR (LOAD) ON DRIVE BUS.	
15	●			PAGE ADR X256, CHECK PAGE ADR ON DRIVE BUS.	
16	●			TALK ADR + 128 TALK DATA. CHECK GROUP ADR (INCM'T) ON DRIVE BUS	
17	●			TALK ADR X128, CHECK GROUP ADR (LOAD) ON DRIVE BUS.	
18	●			TALK DATA X 2^{15}. DATA FROM DRIVE BUS TO CSH.	HLSD/HLSC
19	●			ACC DATA X 2^{15}. DATA FROM CSH TO DRIVE BUS.	
20 21 22		●	PMC SLAVE	WRITE/READ CHECKS OF RAM ON HLMA, 20-INCREMENTAL DATA, 21 & 22 - CHECKERBOARD PATTERNS.	HLMB/HLMA FAILURES ON UNIQUE BIT PATTERNS INDICATE HLMA IS SUSPECT, OTHERWISE, THE HLMB.
23	●			CHECK STA. ERR. DETECTION, REPORT & CLEAR	HLSC/HLSD (SEE NOTE 5)
24	●			CHECK LINK ERR DETECTION, REPORT & CLEAR	SLAVE HLSA/HLSB
25	●			CHECK INTERRUPT REPORT AND ACKNOWLEDGE OF SAME	HLSD/HLSC
26	●			CHECK DRIVE BUS CONTROL LINES - HOLD/READ/WRITE	HLSD

NOTES

1. WHERE TWO PWB'S ARE LISTED, THE FIRST IS THE MOST PROBABLE CAUSE.
2. CABLE CONNECTED TO J3 CONNECTOR ON SLAVE STATION (IF INSTALLED) SHOULD BE DISCONNECTED.
3. TEST NUMBERS ABOVE THOSE LISTED WILL ILLUMINATE ALL FIVE FAULT LIGHTS.
4. TESTS 00 & 01 ARE HTM SELF TESTS - SEE HTM MAINTENANCE INSTRUCTIONS.

5.THIS TEST WILL FAIL ON STATIONS IN WHICH PARITY IS INTENIONALLY DISABLED.

CAUTION

DO NOT TURN OFF POWER SWITCH ON PMC CONSOLE SINCE IT MAY REQUIRE SYSTEM TO BE RELOADED.

(2) Disconnect the JRI coaxial cable (if used), and disconnect the JPO and JPI coaxial cables from the HLXA. See Figure 2-6 for location of PWB's to ensure test is made on loop where problem is suspected.

(3) If HLXB address jumpers are set for "0", change to some other address per Figure 2-7. Record original address.

(4) Set HLXA toggle switch in bypass (center) position.

(5) Set HLMB jumper in "store" position.

e. Set HTM switches as follows:

Switch	Position/Condition
DEMAND ON/OFF	OFF
LOOP OPEN/CLOSE	CLOSE

f. Disable the stall timer on the PMC by jumpering TP1 to TP2 on HDIC PWB.

g. Turn PS5 POWER switch to ON.

h. Verify that top LED on HLMA is illuminated. This indicates that station is in master mode.

i. The IMM routine aborts any running program and turns on the CIAD program. The routine will be entered as indicated below, depending on the IMM status.

Keyboard	Display
RETURN key	IMM aborted
RETURN key	*
RUN, CIAD	* RUN, CIAD
RETURN key	#

j. The IMM test will put the master station in the slave mode. The top LED on the HLMA will light when the station is in the master mode.

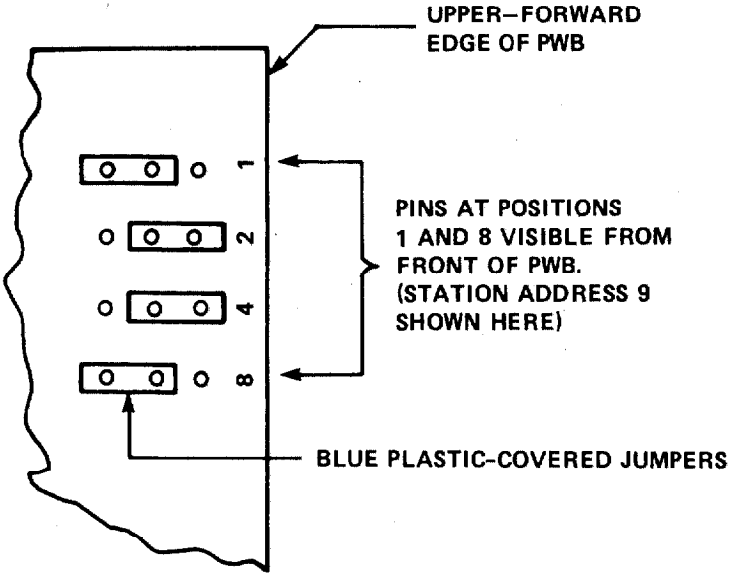


Figure 2-7. Jumpering for Station Address

Keyboard	Display	Failure
MTEST, X, 1	#MTEST, X, 1	
(X=J, K or L depending upon the loop under test).		
RETURN key	Status Error	HLMB
	Verify Error	HLMA
	#	Proceed to

k. If failure occurred during test, replace the suspected PWB, then turn PS5 off for 5 seconds and then back on. Verify that top HLMA LED is on and repeat test.

l. Verify that top HLMA LED is off at the PMC. If on, replace HLMB and repeat step j.

m. Select the slave mode by depressing the HTM SLAVE MODE switch.

n. Set the HTM station address thumbwheel to HLXB setting.

aa. If any indication in step z is incorrect, replace the HLMA.

ab. Set HTM LOOP switch to CLOSE.

ac. Display IMM message as follows:

Keyboard	Display	Failure
RETURN key	*	
BREAK, CIAD	Additional message followed by a #	
MTEST, X, Y	#MTEST, X, Y	

RETURN key #

X = loop number (J, K or L)

Y = 3 through 7 to run tests 3-7 sequentially. Do not enter each successive test unless a # is displayed with no error messages.

NOTE

If an error message occurs on any test, the probable failures are:

Status error - HLMB

Interrupt - HLMB

Verification - HLMA

ad. Depress ABORT key on IMM keyboard. Verify that display blanks.

ae. Reconnect HLMB and HLXB jumpers on the PMC to original positions.

af. Disconnect the HTM cables and reconnect the JRI, JPO, and JPI cables to the HLXA if disconnected in step 2.

af. Return the PMC MODE SELECTION switch to its original position. Return HLXA toggle switch to normal (UP) position.

SECTION 3
THEORY OF OPERATION

3-1. PURPOSE OF HTM

The traffic controller with which the HTM is designed to be used, comprises a master station and up to 15 slave stations connected via a high-speed data link. A master station is the station at the head of each loop that formats and controls the traffic on the loop. The basic purposes of the HTM are described below:

a. The HTM determines whether a master station can be substituted for and checks if the data link works when the HTM is substituted for the master station. If a master station is suspected of being faulty, the HTM can be substituted for the master station and a check made to see if the data link works with the HTM.

b. The HTM can substitute for the entire data link, including the master station, and plugged into a slave station using the HTM in lieu of the entire data link. In other words, the slave station is taken out of the loop and connected to the HTM.

Refer to paragraph 1-7 for a description of HTM modes of operation.

3-2. TEST MESSAGE FORMAT

Two kinds of information are contained in the test message transmitted onto the data link by the HTM. This includes slave station addresses and data. Information received back from the slave stations includes 16 bits of data in the field of one message. The command field of the message contains information as to whom the message is destined for and the intent of the message. The HTM is capable of reconstructing all of the messages in the proper formats. There are approximately ten types of messages that involve transfer and status between master and slave.

3-3. DESCRIPTION

The HTM comprises eight printed circuit cards: HGPA, HGPB, HPIA, HHTA, HSQC, HLXA, HLXB, and HRRA. Cards HGPA and HSQC make up a test controller which generate all control signals for the HTM and establish format of test messages generated. Test, monitoring, and simulation functions performed by the HTM are selected by front panel switches. The commands selected are

SECTION 4
HTM MAINTENANCE INSTRUCTIONS

4-1. PERIODIC MAINTENANCE

The HTM requires no periodic maintenance other than indicated in paragraphs 4-2 through 4-8.

4-2. CLEANING. Inspect the exterior of the HTM. Remove dust and other loose dirt with a clean soft cloth.

4-3. VISUAL INSPECTION. Inspect the HTM, prior to each operation, for obvious visual defects such as loose connectors, frayed cables, etc. When the equipment fails to operate properly, check switches and controls for correct setting.

4-4. POWER SUPPLY CHECKS. The DC power supply voltages should be checked periodically. To do so, turn on the POWER switch and check the power supply voltages at the HTM front panel test points. Voltages should be as follows:

Test Point	Required Voltage
P5	+4.75 to +5.25VDC
P15	+15 to +17.2VDC
N15	-15 to -17.2VDC

4-5. ADJUSTMENT OF +5VDC POWER SUPPLY P5. The +5VDC power supply, P5, requires periodic adjustment, if voltage measured at test point P5 is not within tolerance. This is the only adjustment on the HTM. P5 can be adjusted by opening the HTM and adjusting R40 on the power supply for +5V at test point P5.

4-6. SELF-TEST. The unit has a self-test (tests 00 and 01) which should be run prior to each use. Test 00 reads the switches and displays their value on the DATA AND ADDRESS DISPLAY for visual verification. Test 01 is a go/no-go check of transmit/receive capability. The self-test function verifies approximately 90% of the HTM circuitry. In the event that faults are detected during the self-test, fault isolation procedures are included as an extension to the 00 and 01 tests.

(1) Set MESSAGE SELECT thumbwheel (beginning with 00) to select front panel switches to be tested.

(2) Depress applicable front panel switch(es) to select various combinations of numbers.

(3) Check that corresponding numbers appear on affected HTM displays.

NOTE

DATA AND ADDRESS DISPLAY LED's 0 thru 7 = 1 byte, LED's 8 thru F = 1 byte, and all eight LED's in STATION ADDRESS/ MESSAGE TYPE = 1 byte.

4-8. Test 01 - Transmit/Receive Electronics. Conduct test 01 as follows:

a. Connect HTM TRANSMIT and RECEIVE coaxial cables to the two coaxial shorting plugs on the HTM front panel.

b. Set TEST thumbwheel to 01.

c. Momentarily depress TEST START switch and check status of five FAULT LED's and PASS LED. If any of the FAULT LED's light, proceed with the remaining steps in this procedure and refer to trouble-shooting chart, Table 4-2, if abnormal indication observed.

d. Connect dual channel oscilloscope to following HTM test points:

Channel 1 - DATA OUT

Channel 2 - DATA IN

Sync - ALL

e. Using 1:1 scope probes, set scope vertical gain for 2V/cm. and sweep rate for 5 microseconds/cm.

f. The data-bearing portion of the message (see message structure in Figure 4-1) will be rapidly varying, however, some gross faults may be observed.

Table 4-2. HTM Troubleshooting Chart (Continued)

SYMPTOM	PROBABLE	REMEDY
While performing test 00, STATION thumbwheel is ok but 1 of 16 decoded outputs of DATA AND ADDRESS DISPLAY bad.	Faulty HPIA card.	Replace HPIA card.
While performing test 01, oscilloscope cannot be synchronized.	Faulty HHTA card.	Replace HHTA card.
While performing test 01, no data out (sweep only), or sync fill pattern not as shown in patterns in Figure 4-1.	Faulty HHTA card.	Replace HHTA card.
While performing test 01, data in does not follow data out as shown in Figure 4-1.	Faulty HLXA or HLXB card.	Replace HLXA and HLXB cards.
While performing test 01, data in/out patterns do not appear as shown in Figure 4-1.	Faulty HGPB or HHRA card.	Replace HGPB and HHRA cards.

4-10. DISASSEMBLY INSTRUCTIONS

WARNING

ONLY THOSE PERSONNEL QUALIFIED TO SERVICE ELECTRONIC EQUIPMENT SHOULD ATTEMPT TO DISASSEMBLE THE HTM. THE APPLICATION OF POWER TO THE HTM, WHILE DISASSEMBLED, PLACES LINE POWER ON EXPOSED TERMINALS. THEREFORE, PRECAUTION SHOULD BE TAKEN TO AVOID A SHOCK HAZARD. PRIOR TO REMOVING THE HTM FRONT PANEL, DISCONNECT POWER PLUG TO AVOID POSSIBILITY OF SHOCK HAZARD ON TERMINAL WHICH WILL BE EXPOSED.

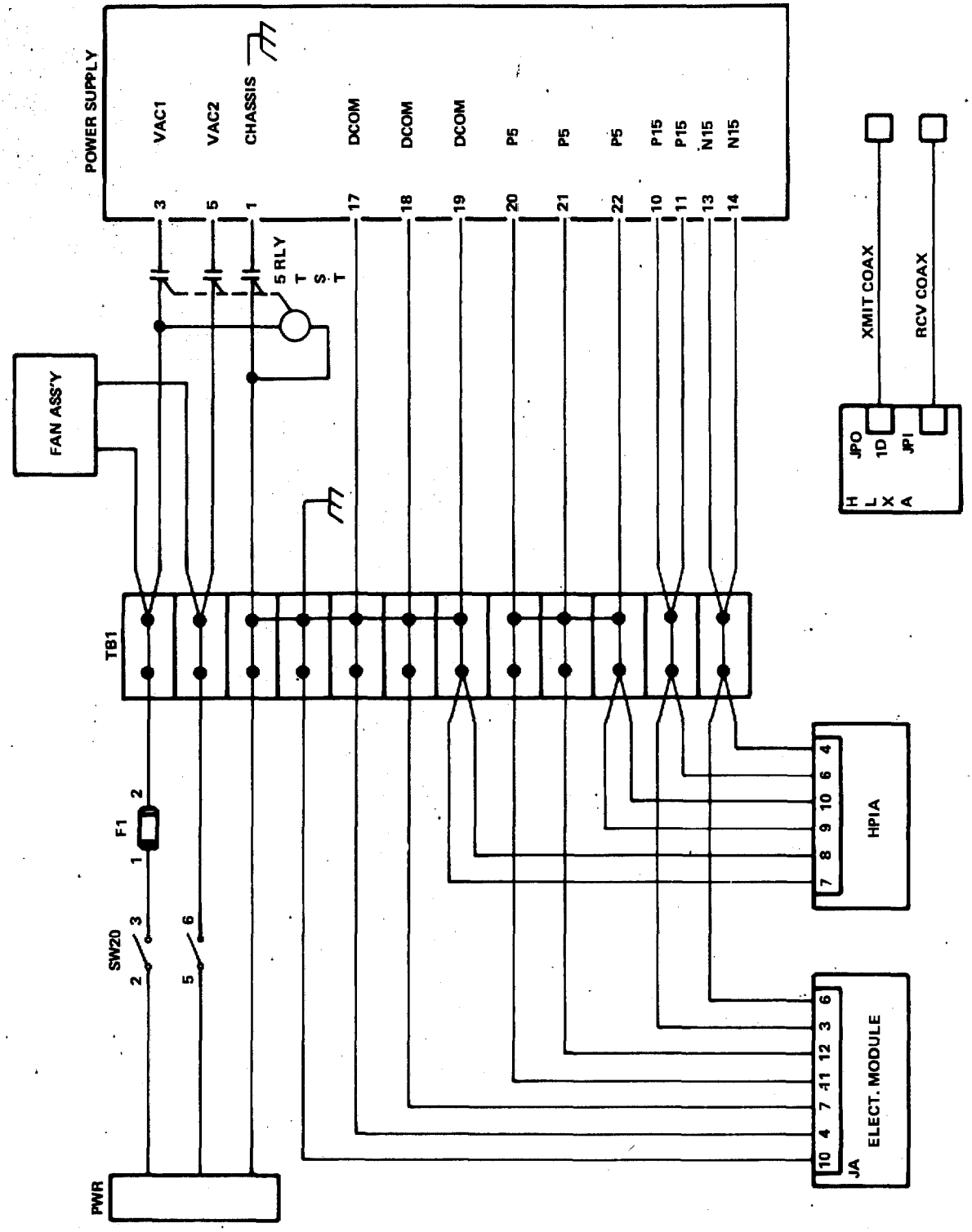


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 1 of 4)

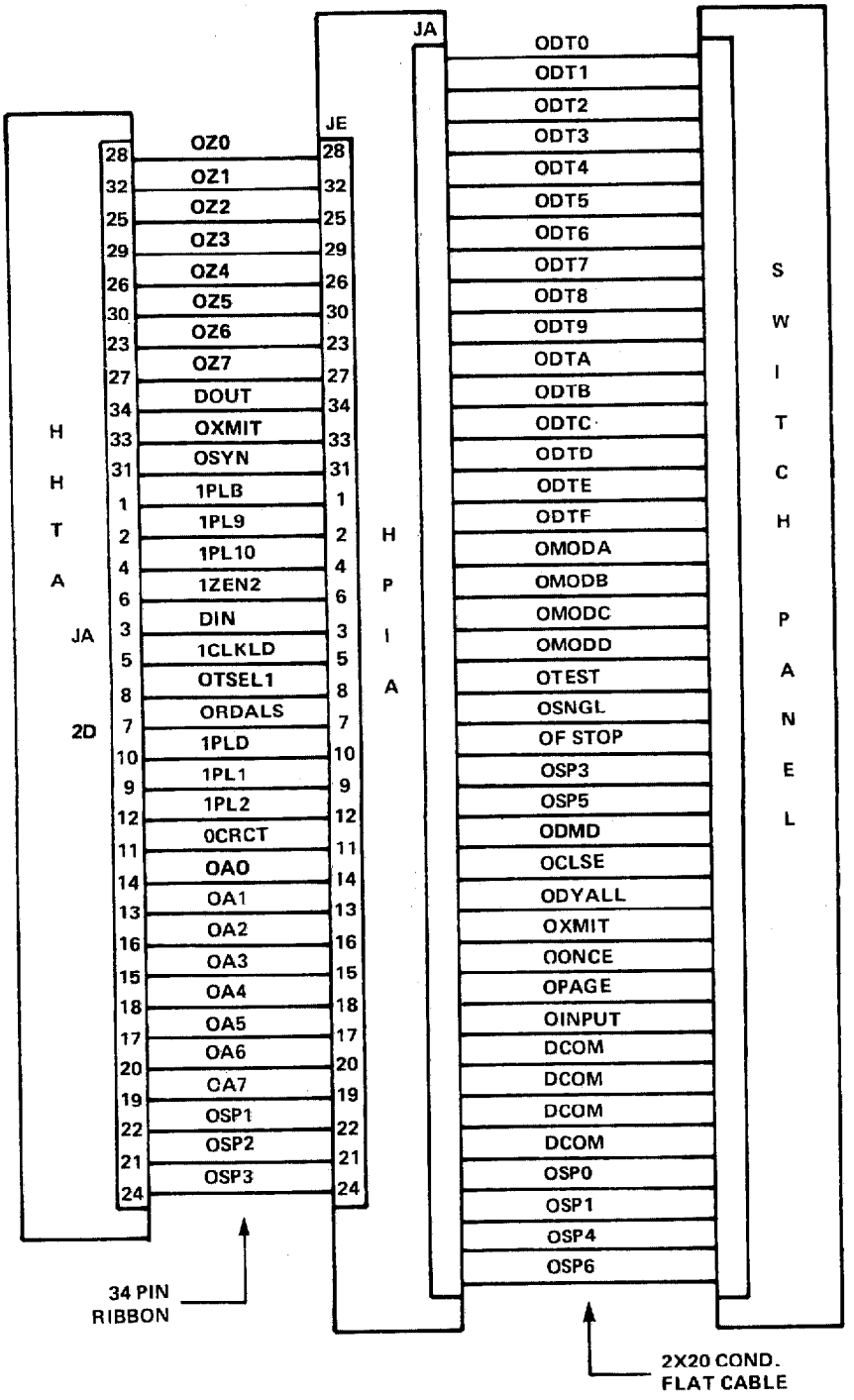


Figure 4-2. Highway Test Module Elementary Wiring Diagram. (Sheet 3 of 4)

SECTION 5
REPLACEMENT PARTS

Table 5-1 is a list of recommended spare parts for the HTM.

Table 5-1. HTM Spare Parts

NOMENCLATURE	CATALOG NUMBER
CARD	DS3800HHRA
CARD	DS3800HLXB
CARD	DS3800HLXA
CARD	DS3800HSQC
CARD	DS3800HGPA
CARD	DS3800HGPB
CARD	DS3800HHTA
FAN	218A4491P5
THUMBWHEEL SWITCH	218A4902P1
THUMBWHEEL SWITCH	218A4902P2
PANEL INTERFACE PWB	DS3800HPIA
POWER SUPPLY	218A4924P1
RIBBON CABLE	DS3815RCC1BL014
RIBBON CABLE	DS3815RCG1BL012
RIBBON CABLE	DS3815RCL1BL003
RIBBON CABLE	DS3815RCC1BL003
SIGNAL CABLE ASM	277A6574G1
COAXIAL CABLE ASM	277A6575G1
COAXIAL CABLE ASM	277A6575G2
FUSE 1.5 AMP	573A392P12
POWER SWITCH	68A7336P11
POWER INTERRUPT RELAY	218A4991P1

APPENDIX

TEST PROGRAM GENERAL INFORMATION

There are two programs which can be run on the IMM; the IMM program and the CIAD program for special tests

IMM PROGRAM

When the IMM program is running, ENTER COMMAND will appear as a prompt on the CRT. This indicates the program is ready for the next input.

CIAD PROGRAM

When the CIAD program is running, the prompt character is a #. The CIAD program is called by entering RUN, CIAD. The CIAD test is selected by entering MTEST, X, Y, C, F, where:

X = loop under test (J, K or L).

Y = test number.

C = continuous and is optional. This causes the test to run repetitively.

F = run continuous without stopping for errors and is optional. This is used for more sophisticated methods using an oscilloscope or logic analyzer.

TERMINATING A TEST

To terminate a test running in continuous mode, depress RETURN key and then enter BREAK, CIAD followed by a RETURN. The IMM will display remaining messages which have been stacked up followed by a # prompt. To terminate the CIAD program, enter ABORT following a # prompt character followed by a RETURN.

(1) Set MESSAGE SELECT thumbwheel (beginning with 00) to select front panel switches to be tested.

(2) Depress applicable front panel switch(es) to select various combinations of numbers.

(3) Check that corresponding numbers appear on affected HTM displays.

NOTE

DATA AND ADDRESS DISPLAY LED's 0 thru 7 = 1 byte, LED's 8 thru F = 1 byte, and all eight LED's in STATION ADDRESS/ MESSAGE TYPE = 1 byte.

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d. Connect dual channel oscilloscope to following HTM test points:

Channel 1 - DATA OUT

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e. Using 1:1 scope probes, set scope vertical gain for 2V/cm. and sweep rate for 5 microseconds/cm.

f. The data-bearing portion of the message (see message structure in Figure 4-1) will be rapidly varying, however, some gross faults may be observed.